

other by 90° , and so are the rotor q and d axes currents in the rotor reference frames. They can be written as

$$I_{qs}^r = I_s \angle \gamma; \quad I_{ds}^r = I_s \angle \gamma + 90^\circ = j I_{qs}^r; \quad I_{qr}^r = I_r \angle \eta; \quad I_{dr}^r = I_r \angle \eta + 90^\circ = j I_{qr}^r$$

where $I_s = 17.08 \text{ A}$; $I_r = 14.69 \text{ A}$; $\gamma = -0.57 \text{ rad}$; $\eta = -3.03 \text{ rad}$.

Using the torque expression shows that $T_e = 0$, even though it is incorrect. In steady state, speed is constant, and therefore the air gap power is proportional to the torque; however, the torque is a function of products of two currents, as seen from the expression derived earlier. The currents are complex variables in steady state, so when products are involved, one of them in each product has to be a complex conjugate, as in the case of the evaluation of power in ac circuits. Hence, the torque expression for use in this case is modified as follows:

$$T_e = \frac{3}{2} \frac{P}{2} L_m (I_{qs}^e \bar{I}_{qs}^e - I_{ds}^e \bar{I}_{qr}^e) \quad (\text{N}\cdot\text{m}) = 25.758 \text{ N}\cdot\text{m}$$

Note the bar on the rotor currents, to indicate that they are complex conjugates of their respective values. This gives the correct value for the torque and matches that from using synchronous reference frames, given in the above. The solution for torque, like that for other output variables, such as the rotor speed and input and output power, is independent of reference frames; it is important to realize that the observation frames cannot affect these machine values. A similar approach is to be used in the stator reference frame model also, for the evaluation of the steady-state electromagnetic torque of the machine: the currents turn out to be complex values.

5.7.8 Equations in Flux Linkages

The dynamic equations of the induction motor in arbitrary reference frames can be represented by using flux linkages as variables. This involves the reduction of a number of variables in the dynamic equations, which greatly facilitates their solution by using analog and hybrid computers. Even when the voltages and currents are discontinuous, the flux linkages are continuous. This gives the advantage of differentiating these variables with numerical stability. In addition, the flux-linkages representation is used in motor drives to highlight the process of the decoupling of the flux and torque channels in the induction and synchronous machines.

The stator and rotor flux linkages in the arbitrary reference frames are defined as

$$\left. \begin{aligned} \lambda_{qs}^c &= L_s i_{qs}^c + L_m i_{qr}^c \\ \lambda_{ds}^c &= L_s i_{ds}^c + L_m i_{dr}^c \\ \lambda_{qr}^c &= L_r i_{qr}^c + L_m i_{qs}^c \\ \lambda_{dr}^c &= L_r i_{dr}^c + L_m i_{ds}^c \end{aligned} \right\} \quad (5.138)$$

The zero-sequence flux linkages are

$$\left. \begin{aligned} \lambda_{os} &= L_{ls} i_{os} \\ \lambda_{or} &= L_{lr} i_{or} \end{aligned} \right\} \quad (5.139)$$

The q axis stator voltage in the arbitrary reference frame is

$$v_{qs}^c = R_s i_{qs}^c + \omega_c (L_s i_{ds}^c + L_m i_{dr}^c) + L_m p i_{qr}^c + L_s p i_{qs}^c \quad (5.140)$$

Substituting from the defined flux-linkages into the voltage equation yields

$$v_{qs}^c = R_s i_{qs}^c + \omega_c \lambda_{ds}^c + p \lambda_{qs}^c \quad (5.141)$$

Similarly, the stator d axis voltage, the d and q axes rotor voltages, and the zero-sequence voltage equations are derived as

$$v_{ds}^c = R_s i_{ds}^c - \omega_c \lambda_{qs}^c + p \lambda_{ds}^c \quad (5.142)$$

$$v_{os}^c = R_s i_{os}^c + p \lambda_{os}^c \quad (5.143)$$

$$v_{qr}^c = R_r i_{qr}^c + (\omega_c - \omega_r) \lambda_{dr}^c + p \lambda_{qr}^c \quad (5.144)$$

$$v_{dr}^c = R_r i_{dr}^c - (\omega_c - \omega_r) \lambda_{qr}^c + p \lambda_{dr}^c \quad (5.145)$$

$$v_{or}^c = R_r i_{or}^c + p \lambda_{or}^c \quad (5.146)$$

These equations can be represented in equivalent circuits.

Power-system engineers and design engineers use normalized or per unit (p.u.) values for the variables. The normalization of the variables is made via reactances rather than inductances. To facilitate such a step, a modified flux linkage is defined whose unit in volts is

$$\psi_{qs}^c = \omega_b \lambda_{qs}^c = \omega_b (L_s i_{qs}^c + L_m i_{qr}^c) = X_s i_{qs}^c + X_m i_{qr}^c \quad (5.147)$$

where ω_b is the base frequency in rad/sec. Similarly the other modified flux linkages are written as

$$\psi_{ds}^c = X_s i_{ds}^c + X_m i_{dr}^c \quad (5.148)$$

$$\psi_{qr}^c = X_r i_{qr}^c + X_m i_{qs}^c \quad (5.149)$$

$$\psi_{dr}^c = X_r i_{dr}^c + X_m i_{ds}^c \quad (5.150)$$

$$\psi_{os}^c = X_s i_{os}^c \quad (5.151)$$

$$\psi_{or}^c = X_r i_{or}^c \quad (5.152)$$

Substituting the flux linkages in terms of the modified flux linkages yields

$$\lambda_{qs}^c = \frac{\psi_{qs}^c}{\omega_b}, \lambda_{ds}^c = \frac{\psi_{ds}^c}{\omega_b}, \lambda_{os}^c = \frac{\psi_{os}^c}{\omega_b}, \lambda_{qr}^c = \frac{\psi_{qr}^c}{\omega_b}, \lambda_{dr}^c = \frac{\psi_{dr}^c}{\omega_b}, \lambda_{or}^c = \frac{\psi_{or}^c}{\omega_b} \quad (5.153)$$

and, by substituting equation (5.153) into equations from (5.141) to (5.146), the resulting equations in modified flux linkages are

$$\begin{aligned} v_{qs}^c &= R_s i_{qs}^c + \frac{\omega_c}{\omega_b} \psi_{ds}^c + \frac{p}{\omega_b} \psi_{qs}^c, v_{ds}^c = R_s i_{ds}^c - \frac{\omega_c}{\omega_b} \psi_{qs}^c + \frac{p}{\omega_b} \psi_{ds}^c, v_{os}^c = R_s i_{os}^c + \frac{p}{\omega_b} \psi_{os}^c, \\ v_{qr}^c &= R_r i_{qr}^c + \frac{(\omega_c - \omega_r)}{\omega_b} \psi_{dr}^c + \frac{p}{\omega_b} \psi_{qr}^c, v_{dr}^c = R_r i_{dr}^c - \frac{(\omega_c - \omega_r)}{\omega_b} \psi_{qr}^c + \frac{p}{\omega_b} \psi_{dr}^c, \\ v_{or}^c &= R_r i_{or}^c + \frac{p}{\omega_b} \psi_{or}^c \end{aligned} \quad (5.154)$$

The electromagnetic torque in flux linkages and currents is derived as

$$T_e = \frac{3}{2} \frac{P}{2} L_m (i_{qs}^c i_{dr}^c - i_{ds}^c i_{qr}^c) = \frac{3}{2} \frac{P}{2} (i_{qs}^c (L_m i_{dr}^c) - i_{ds}^c (L_m i_{qr}^c)) \quad (5.155)$$

The rotor currents can be substituted in terms of the stator currents and stator flux linkages from the basic definitions of the flux linkages. From equation (5.138),

$$L_m i_{dr}^c + L_s i_{ds}^c = \lambda_{ds}^c \quad (5.156)$$

Hence,

$$L_m i_{dr}^c = (\lambda_{ds}^c - L_s i_{ds}^c) \quad (5.157)$$

Similarly,

$$L_m i_{qr}^c = (\lambda_{qs}^c - L_s i_{qs}^c) \quad (5.158)$$

Substituting equations (5.157) and (5.158) into equation (5.152) gives the electromagnetic torque in stator flux linkages and stator currents as

$$T_e = \frac{3}{2} \frac{P}{2} (i_{qs}^c (\lambda_{ds}^c - L_s i_{ds}^c) - i_{ds}^c (\lambda_{qs}^c - L_s i_{qs}^c)) = \frac{3}{2} \frac{P}{2} (i_{qs}^c \lambda_{ds}^c - i_{ds}^c \lambda_{qs}^c) \quad (5.159)$$

Alternatively, the electromagnetic torque in terms of modified flux linkages and currents is

$$T_e = \frac{3}{2} \frac{P}{2} \frac{1}{\omega_b} (i_{qs}^c \psi_{ds}^c - i_{ds}^c \psi_{qs}^c) \quad (5.160)$$

The electromagnetic torque can be expressed in only rotor variables and in many other forms. Note that they are all derivable from equation (5.152) by proper substitutions.

5.7.9 Per-Unit Model

The normalized model of the induction motor is derived by defining the base variables both in the abc and the dqo variables. In the abc frames, let the rms values of the rated phase voltage and current form the base quantities, given as

$$\text{Base Power} = P_b = 3V_{b3}I_{b3} \quad (5.161)$$

where V_{b3} and I_{b3} are three-phase base voltage and current, respectively. Selecting the base quantities in dq frames denoted by V_b and I_b to be equal to the peak value of the phase voltage and current in abc frames, we get

$$V_b = \sqrt{2}V_{b3} \quad (5.162)$$

$$I_b = \sqrt{2}I_{b3} \quad (5.163)$$

Hence, the base power is defined as

$$P_b = 3V_{b3}I_{b3} = 3 \frac{V_b}{\sqrt{2}} \frac{I_b}{\sqrt{2}} = \frac{3}{2} V_b I_b \quad (5.164)$$

The arbitrary reference frame model is chosen to illustrate the normalization process. Consider the q axis stator voltage to begin with, which is given in the following:

$$v_{qs}^c = R_s i_{qs}^c + \omega_c (L_s i_{ds}^c + L_m i_{dr}^c) + L_m p i_{qr}^c + L_s p i_{qs}^c \quad (5.165)$$

Writing the inductances in terms of reactances at base frequency of ω_b gives

$$v_{qs}^c = R_s i_{qs}^c + \frac{X_s}{\omega_b} p i_{qs}^c + \frac{\omega_c X_s}{\omega_b} i_{ds}^c + \frac{X_m}{\omega_b} p i_{qr}^c + \frac{\omega_c X_m}{\omega_b} i_{dr}^c \quad (5.166)$$

It is normalized by dividing by the base voltage, V_b :

$$v_{qsn}^c = \frac{v_{qs}^c}{V_b} = \frac{R_s}{V_b} i_{qs}^c + \frac{X_s}{\omega_b V_b} p i_{qs}^c + \frac{\omega_c X_s}{\omega_b V_b} i_{ds}^c + \frac{X_m}{\omega_b V_b} p i_{qr}^c + \frac{\omega_c X_m}{\omega_b V_b} i_{dr}^c \quad (5.167)$$

Substituting for base voltage in terms of base current, I_b , and base impedance, Z_b , as follows,

$$V_b = I_b Z_b \quad (5.168)$$

into equation (5.167) yields

$$\begin{aligned} v_{qsn}^c = & \left(\frac{R_s}{Z_b} \right) \left(\frac{i_{qs}^c}{I_b} \right) + \frac{1}{\omega_b} \left(\frac{X_s}{Z_b} \right) p \left(\frac{i_{qs}^c}{I_b} \right) + \left(\frac{\omega_c}{\omega_b} \right) \left(\frac{X_s}{Z_b} \right) \left(\frac{i_{ds}^c}{I_b} \right) \\ & + \frac{1}{\omega_b} \left(\frac{X_m}{Z_b} \right) p \left(\frac{i_{qr}^c}{I_b} \right) + \left(\frac{\omega_c}{\omega_b} \right) \left(\frac{X_m}{Z_b} \right) \left(\frac{i_{dr}^c}{I_b} \right) \end{aligned} \quad (5.169)$$

By defining the normalized parameters and variables in the following manner,

$$\begin{aligned} R_{sn} &= \frac{R_s}{Z_b} \text{ p.u.}, X_{sn} = \frac{X_s}{Z_b} \text{ p.u.}, \omega_{mn} = \frac{\omega_r}{\omega_b} \text{ p.u.}, i_{dsn}^c = \frac{i_{ds}^c}{I_b} \text{ p.u.}, v_{dsn}^c = \frac{v_{ds}^c}{V_b} \text{ p.u.} \\ R_{rn} &= \frac{R_r}{Z_b} \text{ p.u.}, X_{rn} = \frac{X_r}{Z_b} \text{ p.u.}, i_{qsn}^c = \frac{i_{qs}^c}{I_b} \text{ p.u.}, i_{drn}^c = \frac{i_{dr}^c}{I_b} \text{ p.u.}, v_{qrn}^c = \frac{v_{qr}^c}{V_b} \text{ p.u.} \\ X_{mn} &= \frac{X_m}{Z_b} \text{ p.u.}, \omega_{cn} = \frac{\omega_c}{\omega_b} \text{ p.u.}, i_{qrn}^c = \frac{i_{qr}^c}{I_b} \text{ p.u.}, v_{qsn}^c = \frac{v_{qs}^c}{V_b} \text{ p.u.}, v_{drn}^c = \frac{v_{dr}^c}{V_b} \text{ p.u.} \end{aligned} \quad (5.170)$$

and by substituting equations in (5.170) into appropriate places in (5.169), v_{qsn}^c is obtained as,

$$v_{qsn}^c = \left(R_{sn} + \frac{X_{sn}}{\omega_b} p \right) i_{qsn}^c + \omega_{cn} X_{sn} i_{dsn}^c + \frac{X_{mn}}{\omega_b} p i_{qrn}^c + \omega_{cn} X_{mn} i_{drn}^c \quad (5.171)$$

Similarly, the other equations are normalized and given below:

$$\begin{bmatrix} v_{qsn}^c \\ v_{dsn}^c \\ v_{qrn}^c \\ v_{drn}^c \end{bmatrix} = \begin{bmatrix} R_{sn} + \frac{X_{sn}}{\omega_b} p & \omega_{cn} X_{sn} & \frac{X_{mn}}{\omega_b} p & \omega_{cn} X_{mn} \\ -\omega_{cn} X_{sn} & R_{sn} + \frac{X_{sn}}{\omega_b} p & -\omega_{cn} X_{mn} & \frac{X_{mn}}{\omega_b} p \\ \frac{X_{mn}}{\omega_b} p & (\omega_{cn} - \omega_{rn}) X_{mn} & R_{rn} + \frac{X_{rn}}{\omega_b} p & (\omega_{cn} - \omega_{rn}) X_{mn} \\ -(\omega_{cn} - \omega_{rn}) X_{mn} & \frac{X_{mn}}{\omega_b} p & -(\omega_{cn} - \omega_{rn}) X_{rn} & R_{rn} + \frac{X_{rn}}{\omega_b} p \end{bmatrix} \begin{bmatrix} i_{qsn}^c \\ i_{dsn}^c \\ i_{qrn}^c \\ i_{drn}^c \end{bmatrix} \quad (5.172)$$

The electromagnetic torque is,

$$T_e = \frac{3}{2} \frac{P}{2} \frac{1}{\omega_b} (i_{qs}^c \psi_{ds}^c - i_{ds}^c \psi_{qs}^c) \quad (5.173)$$

but the base torque is,

$$T_b = \frac{P_b}{\omega_b} = \frac{P}{2} \frac{P_b}{\omega_b} \quad (5.174)$$

from which the normalized electromagnetic torque is obtained as

$$\begin{aligned} T_{en} = \frac{T_e}{T_b} &= \frac{\frac{3}{2} \frac{P}{2} \frac{1}{\omega_b}}{\frac{P}{2} \frac{P_b}{\omega_b}} (i_{qs}^c \psi_{ds}^c - i_{ds}^c \psi_{qs}^c) = \frac{3/2}{P_b} (i_{qs}^c \psi_{ds}^c - i_{ds}^c \psi_{qs}^c) = \frac{3/2}{(3/2) V_b I_b} (i_{qs}^c \psi_{ds}^c - i_{ds}^c \psi_{qs}^c) \\ &= \left(\frac{i_{qs}^c}{I_b} \right) \left(\frac{\psi_{ds}^c}{V_b} \right) - \left(\frac{i_{ds}^c}{I_b} \right) \left(\frac{\psi_{qs}^c}{V_b} \right) = (i_{qsn}^c \psi_{dsn}^c - i_{dsn}^c \psi_{qsn}^c) \text{ p.u.} \end{aligned} \quad (5.175)$$

where

$$\left. \begin{aligned} \psi_{dsn}^c &= \frac{\psi_{ds}^c}{V_b} \text{ p.u.} \\ \psi_{qsn}^c &= \frac{\psi_{qs}^c}{V_b} \text{ p.u.} \end{aligned} \right\} \quad (5.176)$$

Note that the modified flux linkages, ψ_{qs}^c and ψ_{ds}^c , are in volts, and hence ψ_{dsn}^c and ψ_{qsn}^c become dimensionless. The electromechanical dynamic equation is given by

$$T_e = J \frac{d\omega_m}{dt} + T_l + B\omega_m \quad (5.177)$$

where J is the moment of inertia of motor and load, T_l is the load torque, B is the friction coefficient of the load and motor, and ω_m is the mechanical rotor speed.

Normalizing this equation yields

$$\begin{aligned} T_{en} = \frac{T_e}{T_b} &= \frac{J \frac{d\omega_m}{dt}}{\left(\frac{P_b \times P/2}{\omega_b} \right)} + \frac{T_l}{T_b} + \frac{B\omega_m}{\left(\frac{P/2 \times P_b}{\omega_b} \right)} = \frac{J\omega_b\omega_b}{(P/2)^2 P_b} \frac{d}{dt} \left(\frac{\omega_r}{\omega_b} \right) + T_{ln} + \frac{B\omega_b\omega_b\omega_r}{(P/2)^2 P_b\omega_b} \\ &= \frac{J\omega_b^2}{(P/2)^2 P_b} \frac{d\omega_m}{dt} + T_{ln} + \frac{B\omega_b^2}{(P/2)^2 P_b} \omega_m = 2H p \omega_{rn} + T_{ln} + B_n \omega_{rn} \end{aligned} \quad (5.178)$$

where

$$H = \frac{1}{2} \frac{J\omega_b^2}{P_b(P/2)^2} \quad (5.179)$$

fabricated rotors are hardly ever used. The reason is that a fabricated aluminum rotor is expensive, whereas a die-cast aluminum rotor is inexpensive. The fabricated copper-bar rotor is used in large machines where the aluminum die-cast rotor is not available or in high-inertia loads demanding frequent starts, such as in crushers and shredders. Frequent line-supply starts of the induction machine result in higher inrush currents that are multiple times the rated currents and hence produce more losses and forces capable of dislodging the rotor bars.

The aluminum die-cast rotor construction is used in applications having lower load inertia than recommended by National Electrical Manufacturers Association (NEMA) and not required to meet stall condition or very high starting torques. This rotor type is used prevalently and covers as much as 90% of applications.

Insulation: The stator and rotor with windings on the magnetic circuit are immersed in varnish and heat-treated for drying. The insulation sheets between the slots and coils and on the enameled wires and between turns in the coil consist of insulation materials of different classes, known as A, B, F, and H. The choice depends on the maximum temperature rise permissible for each class. The NEMA MG1 and American National Standards Institution CR50.41 specifications for allowable stator temperature rise for various insulation classes are given in Table 5.1. Service factor is the ratio between the steady-state maximum power output capability and the rated power output of the machine. Because of the uncertainty in sizing some loads, service factors greater than 1 are recommended in field applications. Note that a service factor of 1.15 means that the machine can generate 15% more power than its nameplate rating. A majority of induction machines have class F insulation; motors for servo-drive applications usually need class H insulation. Underutilized machines in nonrigorous applications with practically no overloads and infrequent starts can use lower-class insulations. Higher winding temperatures usually result in transmission of heat to bearings, resulting in failures and frequent replacements. Therefore, higher operating winding temperature is not preferred, and that is the reason for a majority of motors to have an insulation class of F or lower.

Rotor Shaft: The rotor shaft is usually made of forged steel for higher speeds (> 3,600 rpm) and has to conform to sizes recommended for power levels by NEMA or other applications-specific industrial standards such as American Petroleum Institute (API) standard 541.

TABLE 5.1 Standards for stator temperature rise

Motor Rating & Voltage	Method of Temperature Measurement	Maximum Winding Temperature (°C) for various classes of insulation						
		Service Factor = 1				Service Factor = 1.15		
		A	B	F	H	A	B	F
All	Resistance	60	80	105	125	70	90	115
≤ 1500 hp	Embedded Detectors	70	90	115	140	80	100	125
> 1500 hp, ≤ 700 V	Embedded Detectors	65	85	110	135	75	95	120
> 1500 hp, > 700V	Embedded Detectors	50	80	105	125	70	90	115

is known as inertial constant, and the normalized friction constant is

$$B_n = \frac{B\omega_b^2}{P_b(P/2)^2} \quad (5.180)$$

The normalized equations of the induction machine in the arbitrary reference frames are given by equations (5.172), (5.175), and (5.178).

5.8 DYNAMIC SIMULATION

The dynamic simulation of the induction machine is explained in this section. The equations of the induction machine in arbitrary reference frames in p.u. are cast in the state-space form as

$$P_1 pX_1 + Q_1 X_1 = u_1 \quad (5.181)$$

where

$$P_1 = \begin{bmatrix} \frac{X_{sn}}{\omega_b} & 0 & \frac{X_{mn}}{\omega_b} & 0 \\ 0 & \frac{X_{sn}}{\omega_b} & 0 & \frac{X_{mn}}{\omega_b} \\ \frac{X_{mn}}{\omega_b} & 0 & \frac{X_{rn}}{\omega_b} & 0 \\ 0 & \frac{X_{mn}}{\omega_b} & 0 & \frac{X_{rn}}{\omega_b} \end{bmatrix} \quad (5.182)$$

$$Q_1 = \begin{bmatrix} R_{sn} & \omega_{cn}X_{sn} & 0 & \omega_{cn}X_{mn} \\ -\omega_{cn}X_{sn} & R_{sn} & -\omega_{cn}X_{mn} & 0 \\ 0 & (\omega_{cn} - \omega_{rn})X_{mn} & R_{rn} & (\omega_{cn} - \omega_{rn})X_{rn} \\ -(\omega_{cn} - \omega_{rn})X_{mn} & 0 & -(\omega_{cn} - \omega_{rn})X_{rn} & R_{rn} \end{bmatrix} \quad (5.183)$$

$$X_1 = [i_{qsn}^c \ i_{dsn}^c \ i_{qrn}^c \ i_{drn}^c]^t \quad (5.184)$$

$$u_1 = [v_{qsn}^c \ v_{dsn}^c \ v_{qrn}^c \ v_{drn}^c]^t \quad (5.185)$$

The equation (5.178) can be rearranged in the state-space form as follows:

$$pX_1 = P^{-1}(u_1 - Q_1 X_1) \quad (5.186)$$

This can be written as

$$pX_1 = A_1 X_1 + B_1 u_1 \quad (5.187)$$

where $A_1 = -P^{-1}Q_1$ and $B_1 = P^{-1}$, and they are evaluated to be

$$B_1 = P_1^{-1} = \frac{1}{\Delta} \begin{bmatrix} X_{rn} & 0 & -X_{mn} & 0 \\ 0 & X_{rn} & 0 & -X_{mn} \\ -X_{mn} & 0 & X_{sn} & 0 \\ 0 & -X_{mn} & 0 & X_{sn} \end{bmatrix} \quad (5.188)$$

where

$$\Delta = \left(\frac{X_{sn} X_{rn} - X_{mn}^2}{\omega_b} \right) \quad (5.189)$$

$$A_1 = -\frac{1}{\Delta} \begin{bmatrix} R_{sn} X_{rn} & k_1 + \omega_{rn} X_{mn}^2 & -R_{rn} X_{mn} & \omega_{rn} X_{rn} X_{mn} \\ -k_1 - \omega_{rn} X_{mn}^2 & R_{sn} X_{rn} & -\omega_{rn} X_{rn} X_{mn} & -R_{rn} X_{mn} \\ -R_{sn} X_{mn} & -\omega_{rn} X_{mn} X_{sn} & R_{rn} X_{sn} & k_1 - \omega_{rn} X_{sn} \\ \omega_{rn} X_{mn} X_{sn} & -R_{sn} X_{mn} & -k_1 + \omega_{rn} X_{rn} X_{sn} & R_{rn} X_{sn} \end{bmatrix} \quad (5.190)$$

where

$$k_1 = \omega_{cn} (X_{sn} X_{rn} - X_{mn}^2) \quad (5.191)$$

The electromechanical equation is

$$T_{en} = 2H p \omega_{rn} + T_{ln} + B_n \omega_{rn} \quad (5.192)$$

where

$$T_{en} = i_{qsn}^c \psi_{dsn}^c - i_{dsn}^c \psi_{qsn}^c \quad (5.193)$$

The modified flux linkages require additional computation; the torque can be conveniently expressed in terms of the normalized currents as

$$T_{en} = X_{mn} (i_{qsn}^c i_{drn}^c - i_{dsn}^c i_{qrn}^c) \quad (5.194)$$

and hence the electromechanical equation can be written as

$$p \omega_{rn} = \frac{X_{mn}}{2H} (i_{qsn}^c i_{drn}^c - i_{dsn}^c i_{qrn}^c) - \frac{T_{ln}}{2H} - \frac{B_n}{2H} \omega_{rn} \quad (5.195)$$

Solving equations (5.187) and (5.195) by numerical integration gives the solutions for currents and speed from which the torque, power output, and losses are calculated. A flowchart describing the steps in the dynamic simulation is shown in Figure 5.17.

Example 5.6

An induction motor has the following constants and ratings:

200 V, 4 pole, 3 phase, 60 Hz, Y connected

$R_s = 0.183 \, \Omega$, $R_r = 0.277 \, \Omega$, $L_m = 0.0538 \, \text{H}$

$L_{\sigma s} = 0.0553 \, \text{H}$, $L_r = 0.056 \, \text{H}$, $B = 0$

Load torque = $T_l = 0 \, \text{N} \cdot \text{m}$, $J = 0.0165 \, \text{kg} \cdot \text{m}^2$

Base power is 5 hp

The motor is at standstill. A set of balanced three-phase voltages at 70.7% of rated values at 60 Hz is applied. Plot the starting characteristics of speed vs. torque, instantaneous stator and rotor currents in all the three reference frames, and stator flux linkages and flux phasor.

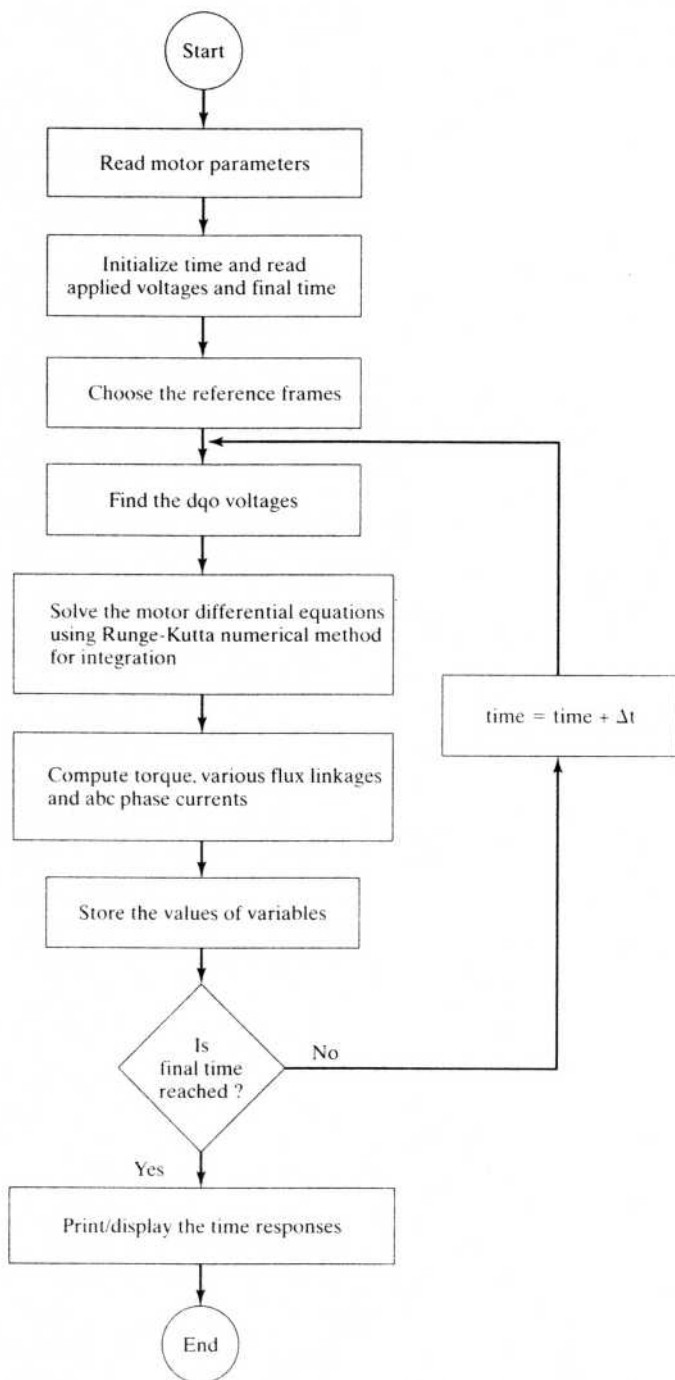


Figure 5.17 Flowchart for dynamic simulation of the induction motor

Solution The plots are shown in Figures 5.18(a), (b) and (c). The machine outputs, such as air gap torque, speed, actual stator and rotor phase currents, and magnitude of stator flux linkages are all the same regardless of reference frames. Also note that the flux linkages are in real units, while other variables are in normalized units. Line start produces higher currents, flux linkages, and dc offsets in them. The torque pulsations are very severe, and repeated line starting could endanger the mechanical integrity of the motor. Higher stator and rotor currents also produce resistive losses that are multiple times the design limit.

5.9 SMALL-SIGNAL EQUATIONS OF THE INDUCTION MACHINE

5.9.1 Derivation

The electrical equations of the induction machine and the electromechanical subsystems given in (5.187), (5.194), and (5.195) combine to give the dynamic equations of the motor-load system. These dynamic equations are nonlinear: some of the terms are the products of two current variables or a current variable and rotor speed. The transient responses or the solution of the dynamic equations are obtained by numerical integration. While any of the standard subroutines can be used for numerical integration, the fourth-order Runge–Kutta method is adequate for the study of induction machines. For the solution, the equations have to be arranged in the state-space form. For controller design with linear control-system design techniques, the nonlinear dynamic equations cannot be directly used. They have to be linearized around an operating point by using perturbation techniques. For small-signal inputs or disturbances, the linearized equations are valid. The ideal model for perturbation to get the linearized model is the one with steady-state operating-state variables as dc values. This is possible only with the synchronously rotating reference frames model of the induction motor. The linearized equations are obtained as follows.

The voltages, currents, torque, stator frequency, and rotor speed in their steady state are designated by an additional subscript with an ‘o’ in the variables, and the perturbed increments are designated by a δ preceding the variables. Accordingly, the variables in SI units after perturbation are

$$v_{qs}^e = v_{qso}^e + \delta v_{qs}^e \quad (5.196)$$

$$v_{ds}^e = v_{dso}^e + \delta v_{ds}^e \quad (5.197)$$

$$i_{qs}^e = i_{qso}^e + \delta i_{qs}^e \quad (5.198)$$

$$i_{ds}^e = i_{dso}^e + \delta i_{ds}^e \quad (5.199)$$

$$i_{qr}^e = i_{qro}^e + \delta i_{qr}^e \quad (5.200)$$

$$i_{dr}^e = i_{dro}^e + \delta i_{dr}^e \quad (5.201)$$

$$T_e = T_{eo} + \delta T_e \quad (5.202)$$

$$T_l = T_{lo} + \delta T_l \quad (5.203)$$

$$\omega_s = \omega_{so} + \delta \omega_s \quad (5.204)$$

$$\omega_r = \omega_{ro} + \delta \omega_r \quad (5.205)$$

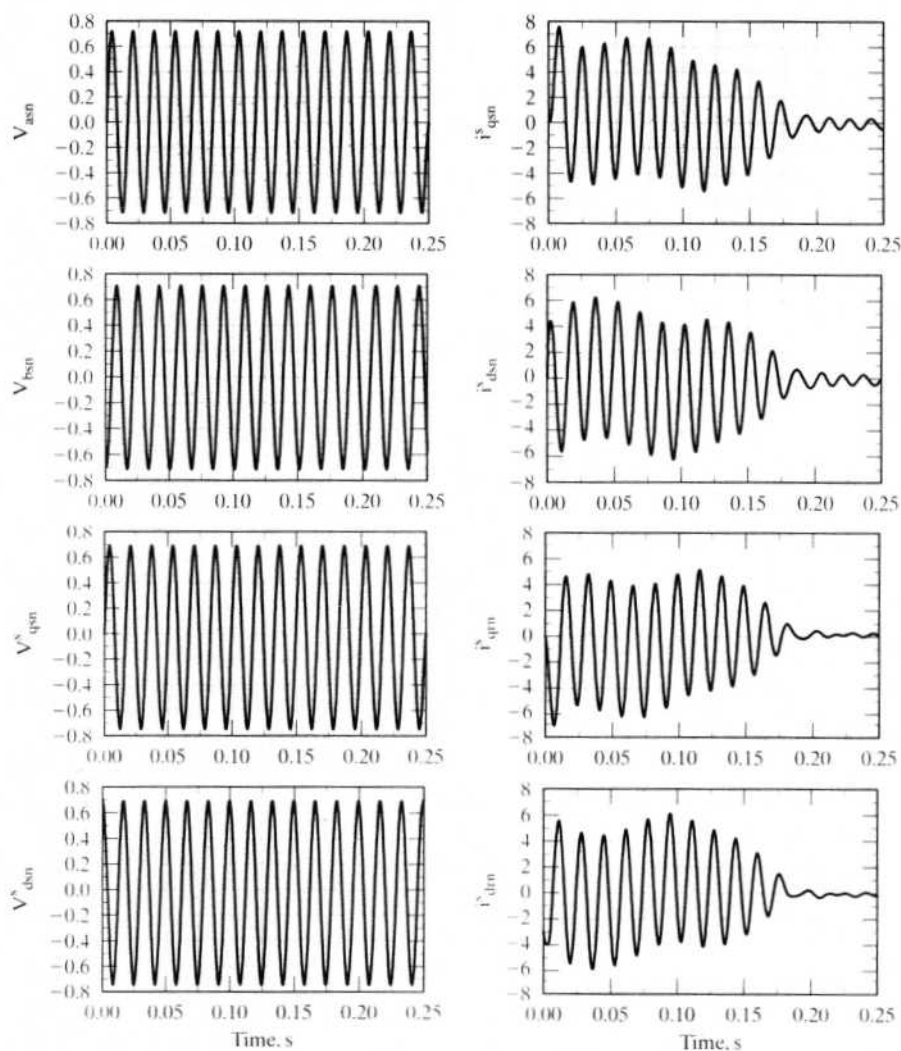


Figure 5.18 (a) Free-acceleration characteristics in stator reference frames (Part I)

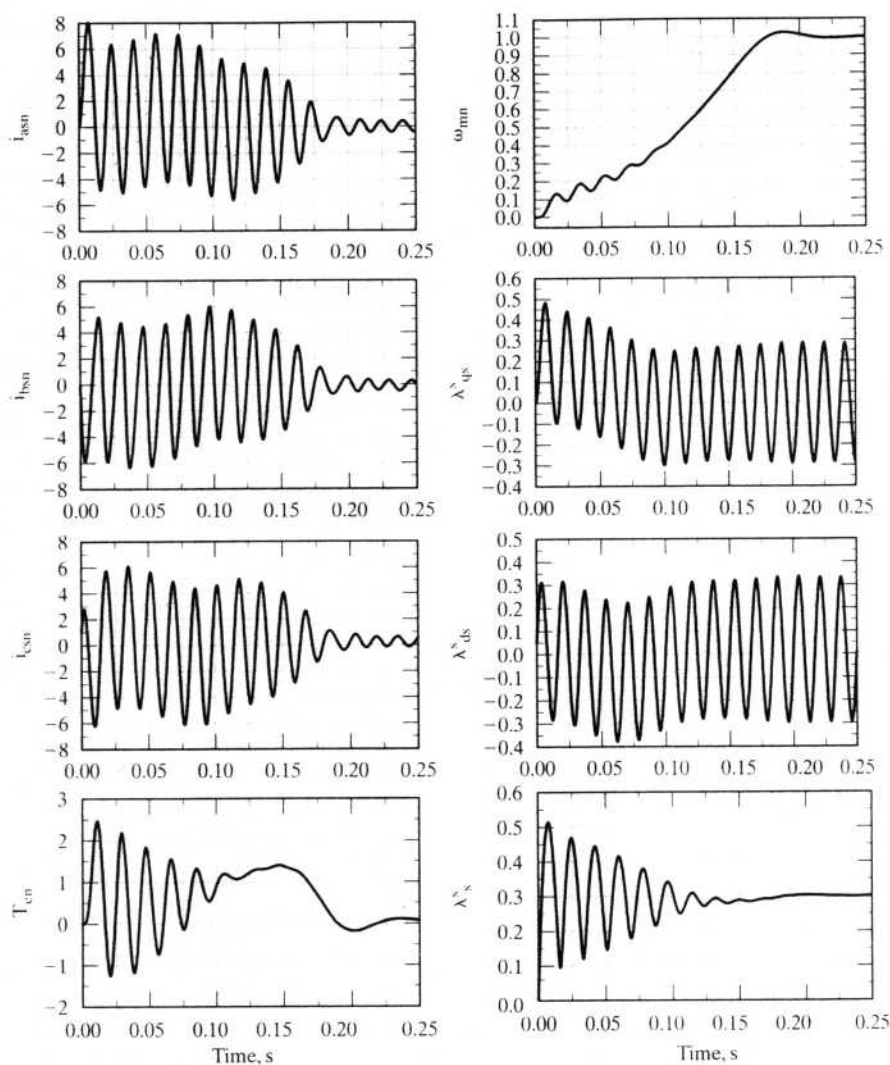


Figure 5.18 (a) Free-acceleration characteristics in stator reference frames (Part II)

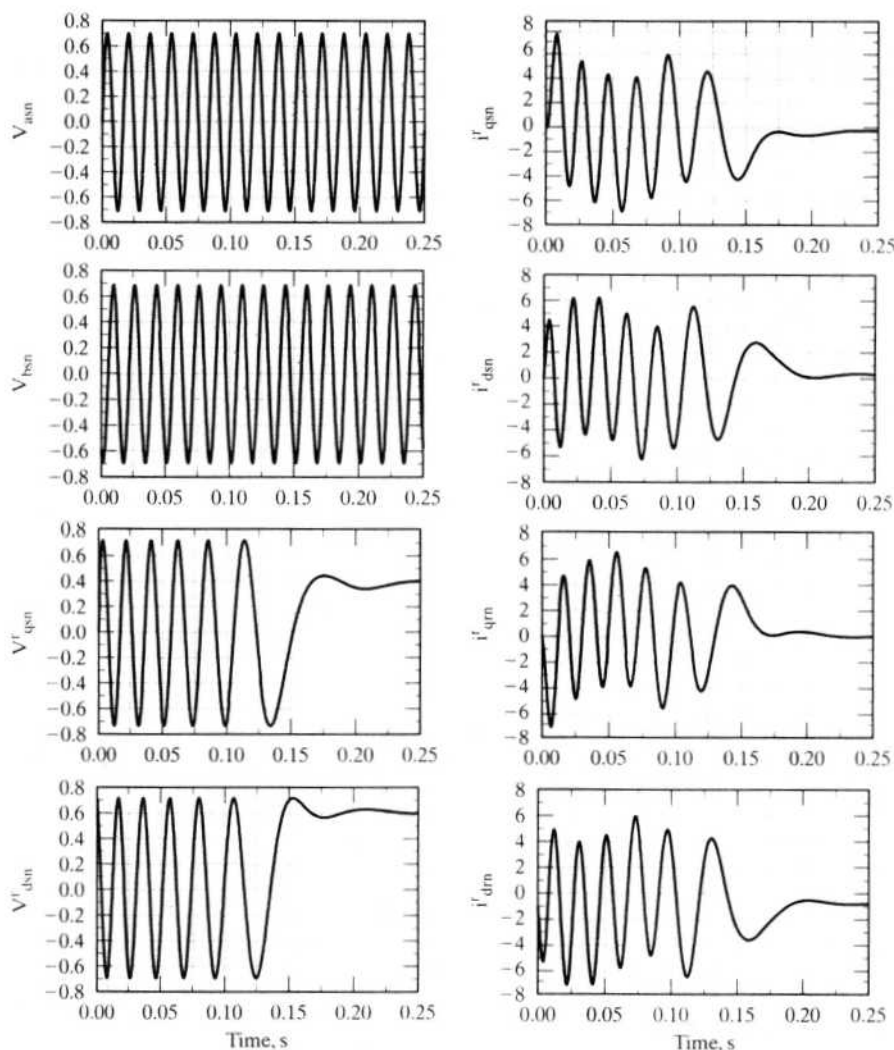


Figure 5.18 (b) Free-acceleration characteristics in rotor reference frames (Part I)

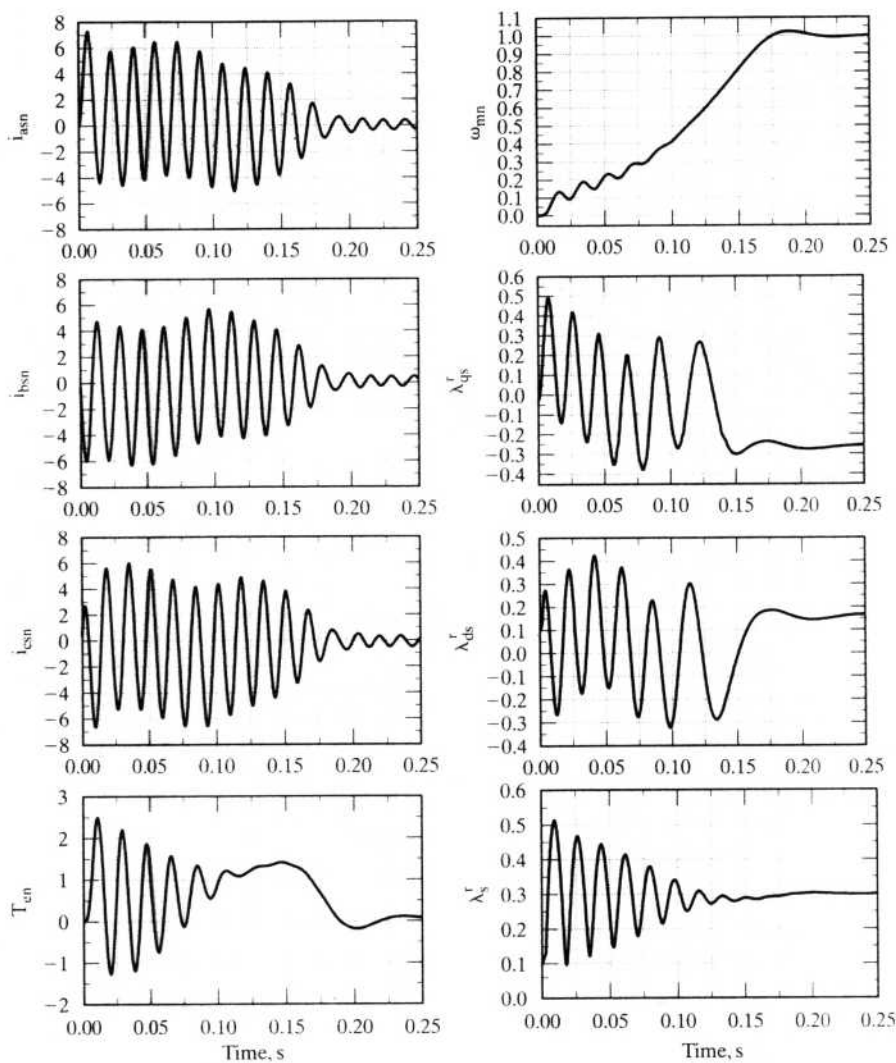


Figure 5.18 (b) Free-acceleration characteristics in rotor reference frames (Part II)

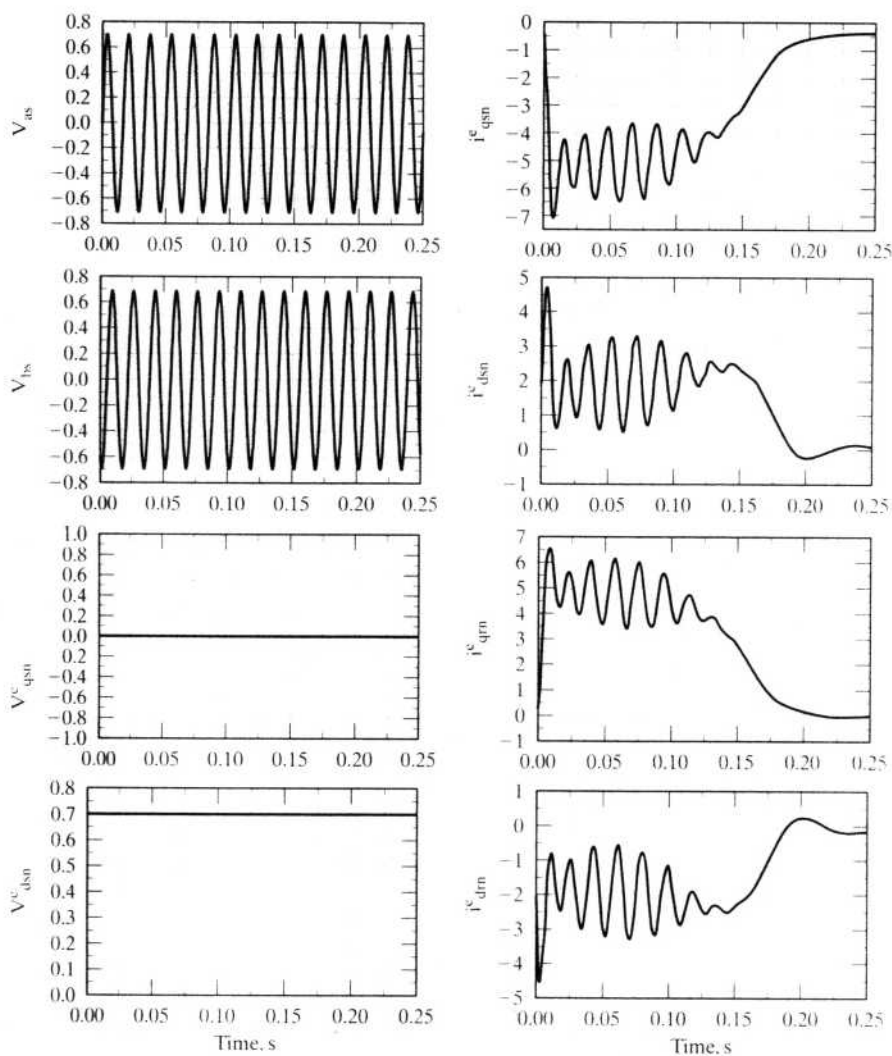


Figure 5.18 (c) Free-acceleration characteristics in synchronous reference frames (Part I)

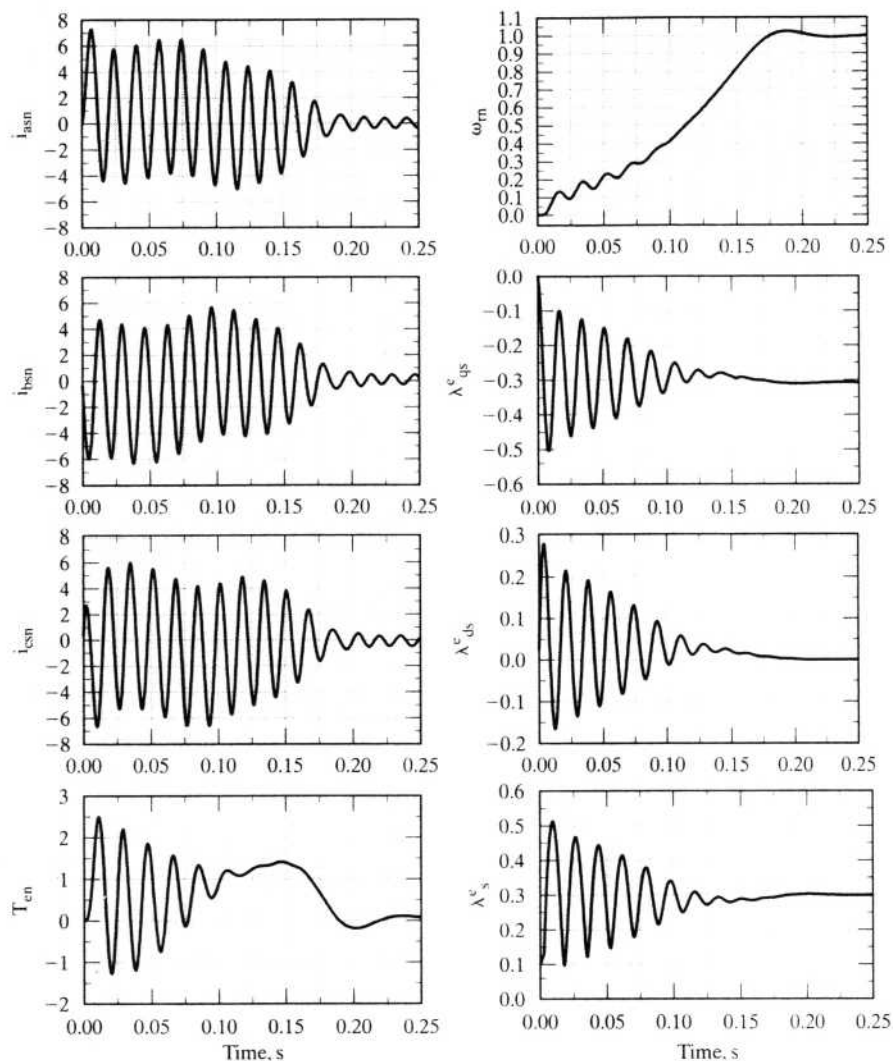


Figure 5.18 (c) Free-acceleration characteristics in synchronous reference frames (Part II)

Enclosure: Windings are inserted in the stator laminations and the stator laminations are fitted inside of a nonmagnetic steel housing. Two end covers will be attached to the steel housing with part of the bearings attached to them. They then will be assembled with through bolts after inserting the rotor with shaft. The non-magnetic steel housing is intended for protecting the stator and rotor assembly from environmental factors (such as rain, water, snow, insects, birds, and rodents making an ingress into the machine), for providing an intermediary medium for exchange of heat between the machine windings and ambient, and for providing mechanical strength and ease of assembly. Commonly used enclosures are open drip-proof (ODP), weather-protected types I and II (WPI, WP II), totally enclosed air-to-air cooled (TEAAC), totally enclosed fan cooled (TEFC), and totally enclosed water-to-air cooled (TEWAC) types. These provide varying degrees of protection from environment and come with cost differentials among them. The enclosures are briefly summarized in the following.

ODP: It will resist the entrance of water that falls at an angle less than 15° from the vertical. It is widely used, but it is not suited for an outdoor application or a dirt-filled environment.

WPI: It is an ODP with the additional screens or louvers to prevent the entrance of objects with diameter greater than 0.75" (1-gm). It is not ideally suited for harsh outdoor application or persistent pollution with airborne materials or abrasive dust. Examples are in textiles manufacturing and mines.

WP II: This is suitable for outdoor applications and can prevent windblown rain or snow. The heavier particles in the air are caught with blow-throughs which are 90° bends creating a low velocity area where the heavier particles drop out before they reach inner electrical parts of the machine. These can also be provided with filters to eliminate particles greater than 10 microns in size. Periodic cleaning and replacement of filters becomes necessary in this situation but comes with the advantage of longer life for the motor.

Totally Enclosed Motors: These enclosures offer the greatest protection against dirt and environment. TEAAC and TEWAC have higher power density compared to TEFC machines. These enclosures are expensive compared to other types. TEAAC has heat pipes, also known as cooling tubes or heat exchangers, that increase the overall volume of the machine with their mounting on top of the machine and also carry the disadvantage of periodic maintenance of the cooling tubes, which could clog. TEWAC has the disadvantage of requiring a clean water supply and auxiliary pump for the operation of its cooling system. It has the advantage of being practically independent of local ambient conditions.

Industrial-grade motors that are not operated continuously should be equipped with a drain, a breather, and a space heater to prevent condensation of moist air due to the breathing in of air that takes place with all types of enclosures.

Bearings: With the proper enclosure chosen for an application, the bearing selection and maintenance is the next step. Between the shaft and the end covers of the stator structure, one bearing on each end is mounted. Two kinds of bearings are available: sleeve and antifriction bearings. Antifriction bearings are cheaper

where

$$T_{eo} = \frac{3}{2} \frac{P}{2} L_m (i_{qso}^e i_{dro}^e - i_{dso}^e i_{qro}^e) \quad (5.206)$$

By substituting from equations (5.196) to (5.206) into equations (5.114), (5.122), and (5.177), by neglecting the second-order terms, and by canceling the steady-state terms on the right- and left-hand sides of the equations, the small-signal dynamic equations are obtained:

$$\delta v_{qs}^e = (R_s + L_s p) \delta i_{qs}^e + \omega_{so} L_s \delta i_{ds}^e + L_m p \delta i_{qr}^e + \omega_{so} L_m \delta i_{dr}^e + (L_s i_{dso}^e + L_m i_{dro}^e) \delta \omega_s \quad (5.207)$$

$$\delta v_{ds}^e = -\omega_{so} L_s \delta i_{qs}^e + (R_s + L_s p) \delta i_{ds}^e - \omega_{so} L_m \delta i_{qr}^e + L_m p \delta i_{dr}^e - (L_s i_{qso}^e + L_m i_{qro}^e) \delta \omega_s \quad (5.208)$$

$$\delta v_{qr}^e = L_m p \delta i_{qs}^e + (\omega_{so} - \omega_{ro}) L_m \delta i_{ds}^e + (R_r + L_r p) \delta i_{qr}^e + (\omega_{so} - \omega_{ro}) L_r \delta i_{dr}^e - (L_m i_{dso}^e + L_r i_{dro}^e) \delta \omega_r + (L_m i_{dso}^e + L_r i_{dro}^e) \delta \omega_s \quad (5.209)$$

$$\delta v_{dr}^e = -(\omega_{so} - \omega_{ro}) L_m \delta i_{qs}^e + L_m p \delta i_{ds}^e - (\omega_{so} - \omega_{ro}) L_r \delta i_{qr}^e + (R_r + L_r p) \delta i_{dr}^e + (L_m i_{qso}^e + L_r i_{qro}^e) \delta \omega_r - (L_m i_{qso}^e + L_r i_{qro}^e) \delta \omega_s \quad (5.210)$$

$$J p \delta \omega_r + B \delta \omega_r = \frac{P}{2} (\delta T_e - \delta T_1) \quad (5.211)$$

$$\delta T_e = \frac{3}{2} \frac{P}{2} L_m (i_{qso}^e \delta i_{dr}^e + i_{dro}^e \delta i_{qs}^e - i_{dso}^e \delta i_{qr}^e - i_{qro}^e \delta i_{ds}^e) \quad (5.212)$$

Combining equations from (5.207) to (5.211) and casting them in state-space form gives

$$pX = AX + B_1 U \quad (5.213)$$

where

$$X = [\delta i_{qs}^e \quad \delta i_{ds}^e \quad \delta i_{qr}^e \quad \delta i_{dr}^e \quad \delta \omega_r]^T \quad (5.214)$$

$$U = [\delta v_{qs}^e \quad \delta v_{ds}^e \quad \delta v_{qr}^e \quad \delta v_{dr}^e \quad \delta \omega_s \quad \delta T_1] t \quad (5.215)$$

$$A = P_1^{-1} Q_1 \quad (5.216)$$

$$B_1 = P_1^{-1} R_1 \quad (5.217)$$

$$P_1 = \begin{bmatrix} L_s & 0 & L_m & 0 & 0 \\ 0 & L_s & 0 & L_m & 0 \\ L_m & 0 & L_r & 0 & 0 \\ 0 & L_m & 0 & L_r & 0 \\ 0 & 0 & 0 & 0 & J \end{bmatrix} \quad (5.218)$$

$$Q_1 = \begin{bmatrix} -R_s & -\omega_{s0}L_s & 0 & -\omega_{s0}L_m & 0 \\ \omega_{s0}L_s & -R_s & \omega_{s0}L_m & 0 & 0 \\ 0 & -(\omega_{s0} - \omega_{r0})L_m & -R_r & -(\omega_{s0} - \omega_{r0})L_r & L_m i_{ds0}^e + L_r i_{dr0}^e \\ (\omega_{s0} - \omega_{r0})L_m & 0 & (\omega_{s0} - \omega_{r0})L_r & -R_r & -(L_m i_{qs0}^e + L_r i_{qr0}^e) \\ k_2 i_{dro}^e & -k_2 i_{qro}^e & -k_2 i_{dso}^e & k_2 i_{qso}^e & -B \end{bmatrix} \quad (5.219)$$

$$k_2 = \frac{3}{2} \left(\frac{P}{2} \right)^2 L_m \quad (5.220)$$

$$R_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & -L_s i_{ds0}^e + L_m i_{dro}^e & 0 \\ 0 & 1 & 0 & 0 & L_s i_{qs0}^e + L_m i_{qro}^e & 0 \\ 0 & 0 & 1 & 0 & -L_m i_{ds0}^e + L_r i_{dro}^e & 0 \\ 0 & 0 & 0 & 1 & L_m i_{qs0}^e + L_r i_{qro}^e & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{P}{2} \end{bmatrix} \quad (5.221)$$

The output of interest can be a function of the state variables and inputs, expressed as

$$y = CX + DU \quad (5.222)$$

where C and D are now vectors of appropriate dimensions. The system and its output are described by the equations (5.213) and (5.222), respectively.

5.9.2 Normalized Small-Signal Equations

The small-signal equations for the normalized model can be derived as in the procedure adopted for the SI-unit model. The equations thus obtained are presented in this section. The small-signal equations for the torque and voltages are given as follows:

$$\delta T_{en} = X_{mn} (i_{qsno}^e \delta i_{drn}^e + i_{drno}^e \delta i_{qsn}^e - i_{dsno}^e \delta i_{qrn}^e - i_{qrno}^e \delta i_{dsn}^e) \quad (5.223)$$

$$\begin{aligned} \delta v_{qsn}^e = & \left(R_{sn} + \frac{X_{sn}}{\omega_b} p \right) \delta i_{qsn}^e + \omega_{sno} X_{sn} \delta i_{dsn}^e + \frac{X_{mn}}{\omega_b} p \delta i_{qrn}^e + \omega_{sno} X_{mn} \delta i_{drn}^e \\ & + (X_{sn} i_{dsno}^e + X_{mn} i_{drno}^e) \delta \omega_{sn} \end{aligned} \quad (5.224)$$

$$\begin{aligned} \delta v_{dsn}^e = & -\omega_{sno} X_{sn} \delta i_{qsn}^e + \left(R_{sn} + \frac{X_{sn}}{\omega_b} p \right) \delta i_{dsn}^e - \omega_{sno} X_{mn} \delta i_{qrn}^e + \frac{X_{mn}}{\omega_b} p \delta i_{drn}^e \\ & - (X_{sn} i_{qsno}^e + X_{mn} i_{qrno}^e) \delta \omega_{sn} \end{aligned} \quad (5.225)$$

$$\begin{aligned} \delta v_{qrn}^e = & \frac{X_{mn}}{\omega_b} p \delta i_{qsn}^e + (\omega_{sno} - \omega_{rno}) X_{mn} \delta i_{dsn}^e + \left(R_{rn} + \frac{X_{rn}}{\omega_b} p \right) \delta i_{qrn}^e \\ & + (\omega_{sno} - \omega_{rno}) X_{rn} \delta i_{drn}^e - (X_{mn} i_{dsno}^e + X_{rn} i_{drno}^e) \delta \omega_{rn} + (X_{mn} i_{dsno}^e + X_{rn} i_{drno}^e) \delta \omega_{sn} \end{aligned} \quad (5.226)$$

$$\begin{aligned} \delta v_{drn}^e = & -(\omega_{sno} - \omega_{rno})X_{mn}\delta i_{qsn}^e + \frac{X_{mn}}{\omega_b}p\delta i_{dsn}^e - (\omega_{sno} - \omega_{rno})X_{rn}\delta i_{qrn}^e \\ & + \left(R_{rn} + \frac{X_{rn}}{\omega_b}p\right)\delta i_{drn}^e + (X_{mn}i_{qsn}^e + X_{rn}i_{qrno}^e)\delta\omega_{rn} - (X_{mn}i_{qsn}^e + X_{rn}i_{qrno}^e)\delta\omega_{sn} \end{aligned} \quad (5.227)$$

$$2Hp\delta\omega_{rn} + B_n\delta\omega_{rn} = (\delta T_{en} - \delta T_{ln}) \quad (5.228)$$

Combining the equations from (5.223) to (5.228) and casting them in state-space form gives

$$px = Ax + B_1u \quad (5.229)$$

where

$$x = [\delta i_{qsn}^e \quad \delta i_{dsn}^e \quad \delta i_{qrn}^e \quad \delta i_{drn}^e \quad \delta\omega_{rn}]^T \quad (5.230)$$

$$u = [\delta v_{qsn}^e \quad \delta v_{dsn}^e \quad \delta v_{qrn}^e \quad \delta v_{drn}^e \quad \delta\omega_{sn} \quad \delta T_{ln}]^T \quad (5.231)$$

$$A = P_1^{-1}Q_1 \quad (5.232)$$

$$B_1 = P_1^{-1}R_1 \quad (5.233)$$

$$P_1 = \begin{bmatrix} \frac{X_{sn}}{\omega_b} & 0 & \frac{X_{mn}}{\omega_b} & 0 & 0 \\ 0 & \frac{X_{sn}}{\omega_b} & 0 & \frac{X_{mn}}{\omega_b} & 0 \\ \frac{X_{mn}}{\omega_b} & 0 & \frac{X_{rn}}{\omega_b} & 0 & 0 \\ 0 & \frac{X_{mn}}{\omega_b} & 0 & \frac{X_{rn}}{\omega_b} & 0 \\ 0 & 0 & 0 & 0 & 2H \end{bmatrix} \quad (5.234)$$

$$Q_1 = \begin{bmatrix} -R_{sn} & -\omega_{sno}X_{sn} & 0 & -\omega_{sno}X_{mn} & 0 \\ \omega_{sno}X_{sn} & -R_{sn} & \omega_{sno}X_{mn} & 0 & 0 \\ 0 & -\omega_{sno}X_{mn} & -R_{rn} & -\omega_{sno}X_{rn} & X_{mn}i_{dsno}^e + X_{rn}i_{drno}^e \\ \omega_{sno}X_{mn} & 0 & \omega_{sno}X_{rn} & -R_{rn} & X_{mn}i_{qsn}^e + X_{rn}i_{qrno}^e \\ X_{mn}i_{drno}^e & -X_{mn}i_{qrno}^e & -X_{mn}i_{dsno}^e & X_{mn}i_{qsn}^e & -B_n \end{bmatrix} \quad (5.235)$$

where

$$\omega_{sno} = \omega_{sno} - \omega_{rno} = \text{slip speed (p.u.)} \quad (5.236)$$

$$R_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & -X_{sn}i_{dsno}^e + X_{mn}i_{drno}^e & 0 \\ 0 & 1 & 0 & 0 & X_{sn}i_{qsn}^e + X_{mn}i_{qrno}^e & 0 \\ 0 & 0 & 1 & 0 & -(X_{mn}i_{dsno}^e + X_{rn}i_{drno}^e) & 0 \\ 0 & 0 & 0 & 1 & X_{mn}i_{qsn}^e + X_{rn}i_{qrno}^e & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix} \quad (5.237)$$

The output of interest can be a function of the state variables and inputs, expressed as

$$y = Cx + Du \quad (5.238)$$

where C and D are now vectors of appropriate dimensions.

The system and its output are described by the equations (5.229) and (5.238), respectively. They can be used to study the control characteristics of the system, as shown in the next section.

5.10 EVALUATION OF CONTROL CHARACTERISTICS OF THE INDUCTION MACHINE

The control characteristics of the induction machine consist of stability and of frequency and time responses. They require the evaluation of various transfer functions. Stability is evaluated by finding the eigenvalues of the system matrix A in the equation (5.213). It can be found by using standard subroutines available in a software library. The evaluation of transfer functions and of frequency and time responses is available in control-system simulation libraries. Simple algorithms to develop the above are given in this section.

5.10.1 Transfer Functions and Frequency Responses

Taking the Laplace transform of equations (5.213) and (5.222) with the assumption of zero initial conditions gives

$$sX(s) = AX(s) + B_1u(s) \quad (5.239)$$

$$y(s) = CX(s) + Du(s) \quad (5.240)$$

By manipulating equations (5.239) and (5.240), the output can be expressed as

$$y(s) = (C(sI - A)^{-1}B_1 + D)u(s) \quad (5.241)$$

where I is an identity matrix of appropriate dimensions.

The transfer function involves one input, so the input matrix product $B_1u(s)$ is then written as

$$B_1u(s) = b_iu_i(s) \quad (5.242)$$

where b_i is the i^{th} column vector of B matrix and i corresponds to the element number in the input vector. Then, correspondingly,

$$Du(s) = d_iu_i(s) \quad (5.243)$$

Then the resulting equations are

$$\begin{aligned} sX(s) &= AX(s) + b_iu_i(s) \\ y(s) &= CX(s) + d_iu_i(s) \end{aligned} \quad (5.244)$$

The evaluation of transfer functions is made simple if the canonical or phase-variable form of the state equation given in equation (5.244) is found. Assume that is effected by the following transformation,

$$X = T_p X_p \quad (5.245)$$

Then the state and output equations are transformed to

$$pX_p = A_p X_p + B_p u_i \quad (5.246)$$

$$y = C_p X_p + d_i u_i \quad (5.247)$$

where

$$A_p = T_p^{-1} A T_p \quad (5.248)$$

$$B_p = T_p^{-1} b_i \quad (5.249)$$

$$C_p = C T_p \quad (5.250)$$

These matrices and vectors are of the form

$$A_p = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -m_1 & -m_2 & -m_3 & -m_4 & -m_5 \end{bmatrix} \quad (5.251)$$

$$B_p = [0 \ 0 \ 0 \ 0 \ 1]^T \quad (5.252)$$

$$C_p = [n_1 \ n_2 \ n_3 \ n_4 \ n_5] \quad (5.253)$$

and the transfer function, by observation, is written as

$$\frac{y(s)}{u_i(s)} = \frac{n_1 + n_2 s + n_3 s^2 + n_4 s^3 + n_5 s^4}{m_1 + m_2 s + m_3 s^2 + m_4 s^3 + m_5 s^4 + s^5} + d_i \quad (5.254)$$

The problem lies in finding the transformation matrix, T_p . An algorithm to construct T_p is given below:

$$\left. \begin{aligned} T_p &= [t_1 \ t_2 \ t_3 \ t_4 \ t_5] \\ t_5 &= b_i \\ t_{5-k} &= A t_{5-k+1} + m_{5-k+1} b_i; k = 1, 2, 3, 4 \end{aligned} \right\} \quad (5.255)$$

where, $t_1, \dots, t_5$ are the column vectors. The last equation needs the coefficients of the characteristic equation and is computed beforehand by using the Leverrier algorithm. The Leverrier algorithm is given in the following.

$$\left. \begin{aligned} m_5 &= -\text{trace}(A); H_5 = A + m_5 I \\ m_4 &= -\frac{1}{2} \text{trace}(A H_5); H_4 = A H_5 + m_4 I \\ &\vdots \\ m_1 &= -\frac{1}{5} \text{trace}(A H_2) \end{aligned} \right\} \quad (5.256)$$

where the trace of a matrix is equal to the sum of its diagonal elements. The frequency response is evaluated from equation (5.241) by substituting $s = j\omega$ wherever s occurs. The magnitude and phase plots can be drawn over the desired frequency range for the evaluation of control properties. A flowchart for the computation of transfer functions and of frequency and time responses is given in Figure 5.19. The computation of time responses is considered next.

5.10.2 Computation of Time Responses

If the state equation given in (5.213) is transformed into diagonal form by the transformation

$$X = T_d Z \quad (5.257)$$

then the transformed equations are

$$\dot{Z} = T_d^{-1} A T_d Z + T_d^{-1} B_1 U = MZ + HU \quad (5.258)$$

where M is a diagonal matrix with distinct eigenvalues and

$$M = T_d^{-1} A T_d \quad (5.259)$$

$$H = T_d^{-1} B \quad (5.260)$$

Solving for $z_1, \dots, z_n$ from equation (5.260) gives

$$z_n(t) = \left(\sum_{j=1}^m H_{nj} u_j \right) \frac{(-1 + e^{\lambda_n t})}{\lambda_n} \quad (5.261)$$

where λ_n is the n^{th} eigenvalue and m is the number of inputs. Once vector Z is evaluated via equation (5.263), the output y is obtained from the following:

$$y = CX + DU = CT_d Z + DU \quad (5.262)$$

Example 5.7

For the induction motor given in Example 5.4 and for the steady-state operating conditions given below, find the transfer function between the rotor speed and load torque and the poles of the system.

$$\omega_{s0} = 377 \text{ rad/sec}$$

$$\omega_{r0} = 370 \text{ rad/sec}$$

Solution The steady-state current vector is obtained as follows: The impedance matrix in synchronous reference frame is

$$Z = \begin{bmatrix} R_s & L_s \omega_{s0} & 0 & L_m \omega_{s0} \\ -L_s \omega_{s0} & R_s & -L_m \omega_{s0} & 0 \\ 0 & L_m \omega_{s0} & R_r & L_s \omega_{s0} \\ -L_m \omega_{s0} & 0 & -L_s \omega_{s0} & R_r \end{bmatrix}$$

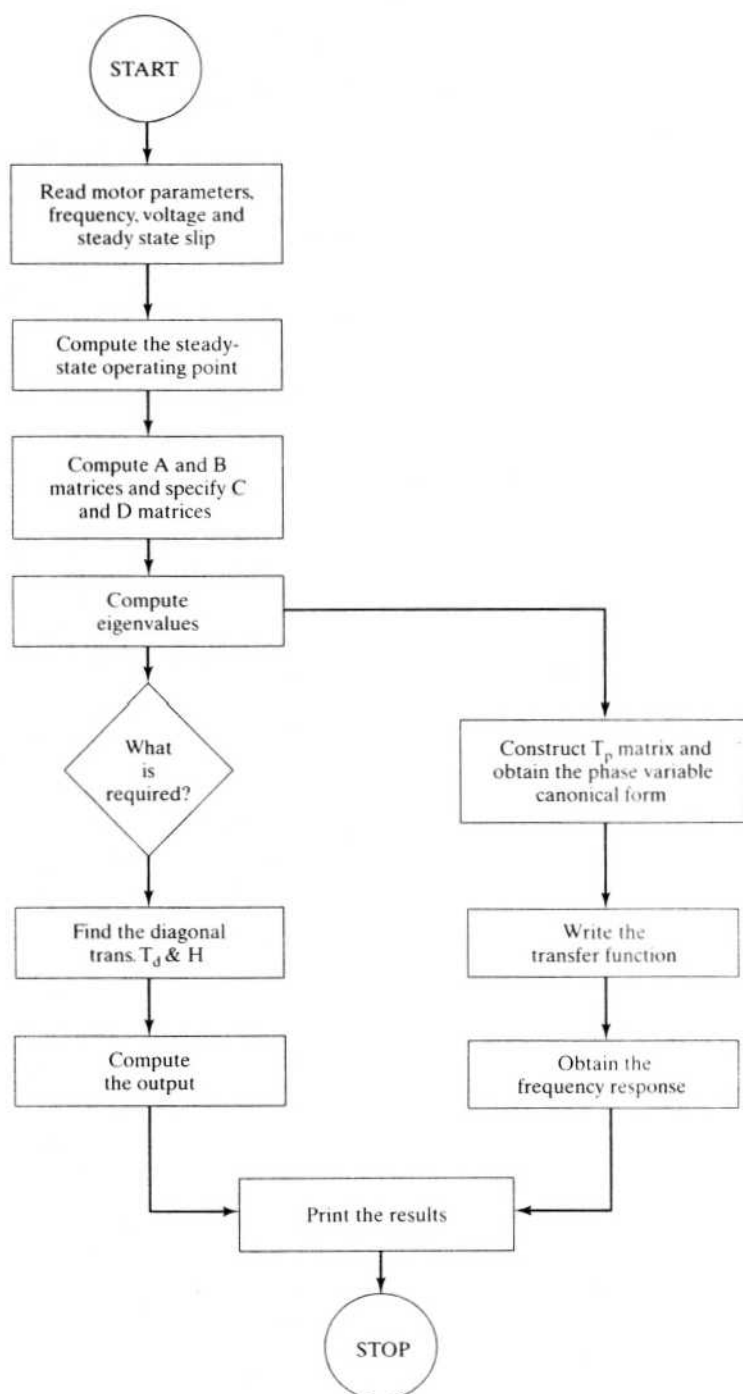


Figure 5.19 Flowchart for the computation of transfer functions and frequency responses

where

$$\omega_{s0} = 377 \text{ rad/sec}$$

$$\omega_{slo} = \omega_{s0} - \omega_{r0} = 37 - 370 = 7 \text{ rad/sec}$$

The q axis stator voltage in synchronous reference frame is

$$v_{qso}^e = 0$$

and

$$v_{dso}^e = \frac{\sqrt{2}}{\sqrt{3}} V = 163.3 \text{ V}$$

$$v = \begin{bmatrix} 0 \\ 163.3 \\ 0 \\ 0 \end{bmatrix}$$

$$I = [Z^{-1}] \cdot v = \begin{bmatrix} -9.48 \\ 15.15 \\ 1.90 \\ -15.44 \end{bmatrix}, A = \begin{bmatrix} I_{qso}^e \\ I_{dso}^e \\ I_{qro}^e \\ I_{dro}^e \end{bmatrix}$$

The steady-state torque is

$$\begin{aligned} T_{co} &= \frac{3}{2} \cdot \frac{P}{2} \cdot L_m (I_{qso}^e I_{dro}^e - I_{dso}^e I_{qro}^e) \\ &= 18.98 \text{ N}\cdot\text{m} \end{aligned}$$

The next step is to compute the system matrices A and B_1 as follows :

$$A = P_1^{-1}Q$$

$$B_1 = P_1^{-1}R_1$$

where

$$P_1 = \begin{bmatrix} L_s & 0 & L_m & 0 & 0 \\ 0 & L_s & 0 & L_m & 0 \\ L_m & 0 & L_r & 0 & 0 \\ 0 & L_m & 0 & L_r & 0 \\ 0 & 0 & 0 & 0 & J \end{bmatrix}$$

$$R_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & -C_1 & 0 \\ 0 & 1 & 0 & 0 & C_2 & 0 \\ 0 & 0 & 1 & 0 & -C_3 & 0 \\ 0 & 0 & 0 & 1 & C_4 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{P}{2} \end{bmatrix}$$

$$Q_1 = \begin{bmatrix} -R_s & -L_s \omega_{so} & 0 & -L_m \omega_{so} & 0 \\ L_s \omega_{so} & -R_s & L_m \omega_{so} & 0 & 0 \\ 0 & -L_m \omega_{slo} & -R_r & -L_r \omega_{slo} & C_3 \\ -L_m \omega_{slo} & 0 & -L_r \omega_{slo} & -R_r & -C_4 \\ K_1 I_{dro}^e & -k_1 I_{qro}^e & -k_1 I_{dso}^e & k_1 I_{qso}^e & -B \end{bmatrix}$$

where

$$C_1 = L_s I_{dso}^e + L_m I_{dro}^e = 0.007$$

$$C_2 = L_s I_{qso}^e + L_m I_{qro}^e = -0.422$$

$$C_3 = L_m I_{dso}^e + L_r I_{dro}^e = -0.05$$

$$C_4 = L_m I_{qso}^e + L_r I_{qro}^e = -0.404$$

$$k_1 = \frac{3}{2} \left(\frac{P}{2} \right)^2 \cdot L_m = 0.323$$

The fifth column in the R_1 matrix is due to the $\delta \omega_s$ term. The output is incremental torque, which is written as the elements of C vector as follows:

$$C = [k_2 I_{dro}^e \quad -k_2 I_{qro}^e \quad -k_2 I_{dso}^e \quad k_2 I_{qso}^e \quad 0]$$

where

$$k_2 = \frac{3}{2} \cdot \frac{P}{2} \cdot L_m$$

Note $D = 0$.

The coefficients of the characteristic polynomial are obtained from Leverrier's algorithm by using the A matrix, as :

$$m_1 = 1.132 \times 10^{11}$$

$$m_2 = 2.706 \times 10^9$$

$$m_3 = 1.646 \times 10^7$$

$$m_4 = 1.75 \times 10^5$$

$$m_5 = 253.63$$

To compute the numerator polynomial, the column vector of the B_1 matrix is identified as corresponding to the input which is the sixth column for the load torque input:

$$b_1 = 6^{\text{th}} \text{ column of } B_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -119.8 \end{bmatrix}$$

By using A , m_1 to m_5 , and b_1 , the column vectors t_5 to t_1 are found via the algorithm

$$t_n = b_1 \quad \text{for } n = 5$$

$$t_{n-1} = A t_n + m_n b_1 \quad \text{for } n = 5, 4, 3, 2$$

The transformation matrix T_p is

$$T_p = [t_1 \ t_2 \ t_3 \ t_4 \ t_5]$$

$$= \begin{bmatrix} -2.31 \times 10^{10} & 1.39 \times 10^8 & -2.91 \times 10^5 & -1.58 \times 10^3 & 0 \\ 9.25 \times 10^{10} & 1.88 \times 10^9 & 1.62 \times 10^6 & 1.29 \times 10^4 & 0 \\ 2.51 \times 10^{10} & -1.17 \times 10^8 & 3.07 \times 10^5 & 1.63 \times 10^3 & 0 \\ -9.48 \times 10^{10} & -1.93 \times 10^9 & -1.73 \times 10^6 & -1.32 \times 10^4 & 0 \\ -4.57 \times 10^{10} & -1.71 \times 10^9 & -1.90 \times 10^7 & -3.03 \times 10^4 & -119.8 \end{bmatrix}$$

and

$$A_p = T_p^{-1} \cdot A \cdot T_p = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -1.13 \times 10^{11} & -2.7 \times 10^9 & -1.65 \times 10^7 & -1.75 \times 10^5 & -253.63 \end{bmatrix}$$

$$C_p = C \cdot T_p = [1.13 \times 10^{11} \ 2.32 \times 10^9 \ 2.12 \times 10^6 \ 1.62 \times 10^4 \ 0] = [n_1 \ n_2 \ n_3 \ n_4 \ n_5]$$

The transfer function then is

$$\frac{\delta T_c(s)}{\delta T_1(s)} = \frac{n_1 + n_2 s + n_3 s^2 + n_4 s^4 + n_5 s^5}{m_1 + m_2 s + m_3 s^2 + m_4 s^4 + m_5 s^4 + s^5}$$

Compute $B_p = T_p^{-1} \cdot b_1$ to check the calculations.

$$B_p = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

proving the correctness of the solution.

5.11 SPACE-PHASOR MODEL

5.11.1 Principle

The stator and rotor flux-linkage phasors are the resultant stator and rotor flux linkages and are found by taking the vector sum of the respective d and q components of the flux linkages. Note that the flux-linkage phasor describes its spatial distribution. Instead of using two axes such as the d and q for a balanced polyphase machine, the flux-linkage phasors can be thought of as being produced by equivalent single-phase stator and rotor windings. Such a representation has many advantages: (i) the system of equations could be compact and be reduced from four to two; (ii) the system reduces to a two-winding system like the dc machine—hence the apparent similarity of them in control to obtain a decoupled independent flux and torque control as in the dc machine; (iii) a clear conceptualization of the dynamics of the machine—it is easier to visualize the interaction of two windings rather than

than sleeve bearings and require lubrication (grease) at periodic intervals, whereas the sleeve bearings need to be checked on a daily basis for oil level and vibration. Further, sleeve bearings cannot take side or vertically upward loads. Assembly and disassembly are much easier with antifriction than with sleeve bearings. Sleeve bearings are not good with underload or overload, whereas the anti friction bearings have a moderate tolerance for both the situations. All of these facts make antifriction bearings a popular choice in a majority of applications.

Rotor Balancing: Before final assembly, the rotor is dynamically balanced so that no eccentricities of any kind are presented to the bearings. The balancing is achieved either by removing some magnetic iron material in smaller motors or by adding a magnetically and electrically inert material compound in larger machines.

For detailed information on the construction details of the machine, refer to the papers and standard texts cited in the reference section of this chapter or to a multitude of textbooks on electrical machines.

5.2.2 Principle of Operation

It is well known that, when a set of three-phase currents displaced in time from each other by angular intervals of 120° is injected into a stator having a set of three-phase windings displaced in space by 120° electrical, a rotating magnetic field is produced. This rotating magnetic field has a uniform strength and travels at an angular speed equal to its stator frequency. It is assumed that the rotor is at standstill. The rotating magnetic field in the stator induces electromagnetic forces in the rotor windings. As the rotor windings are short-circuited, currents start circulating in them, producing a reaction. As known from Lenz's law, the reaction is to counter the source of the rotor currents, i.e., the induced emfs in the rotor and, in turn, the rotating magnetic field itself. The induced emfs will be countered if the difference in the speed of the rotating magnetic field and the rotor becomes zero. The only way to achieve it is for the rotor to run in the same direction as that of the stator magnetic field and catch up with it eventually. When the differential speed between the rotor and magnetic field in the stator becomes zero, there is zero emf, and hence zero rotor currents resulting in zero torque production in the motor. Depending on the shaft load, the rotor will settle down to a speed, ω_r , always less than the speed of the rotating magnetic field, called the *synchronous speed* of the machine, ω_s . The speed differential is known as the *slip speed*, ω_{sl} . The elementary relationships between slip speed, rotor speed, and stator frequency are given below.

Synchronous speed is given as

$$\omega_s = 2\pi f_s, \text{ rad/sec} \quad (5.5)$$

where f_s is the supply frequency.

If ω_m is the mechanical rotor speed, slip speed is

$$\omega_{sl} = \omega_s - \omega_r = \omega_s - \frac{P}{2} \omega_m, \text{ rad/sec} \quad (5.6)$$

four windings, resulting in an in-depth understanding of the dynamic process used in developing high-performance control strategies; (iv) a meaningful interpretation of the eigenvalues of the system, regardless of the reference frames considered in a study; and (v) easier analytical solution of dynamic transients of the key machine variables, involving only the solution of two differential equations with complex coefficients. Such an analytical solution improves the understanding of the machine behavior in terms of machine parameters, leading to the formulation of the machine design requirements for variable-speed applications. Such a space-vector model is derived in this section.

5.11.2 DQ Flux-Linkages Model Derivation

The induction-machine model, in terms of the flux linkages described in section 5.7.8, is made use of in this section. The equations can be cast in the normalized form by finding the currents in terms of the flux linkages from equation (5.138):

$$\left. \begin{aligned} i_{qs}^c &= \frac{1}{\Delta_l} (L_r \lambda_{qs}^c - L_m \lambda_{qr}^c) \\ i_{ds}^c &= \frac{1}{\Delta_l} (L_r \lambda_{ds}^c - L_m \lambda_{dr}^c) \\ i_{qr}^c &= \frac{1}{\Delta_l} (-L_m \lambda_{qs}^c + L_s \lambda_{qr}^c) \\ i_{dr}^c &= \frac{1}{\Delta_l} (-L_m \lambda_{ds}^c + L_s \lambda_{dr}^c) \end{aligned} \right\} \quad (5.263)$$

$$\text{where } \Delta_l = (L_s L_r - L_m^2) \quad (5.264)$$

Then, substituting for the currents in the flux-linkage equations given in (5.141), (5.142), (5.144), and (5.145), the model in arbitrary reference frames in normalized units is derived as

$$\frac{d}{d\tau} \begin{bmatrix} \lambda_{qsn}^c \\ \lambda_{dsn}^c \\ \lambda_{qrn}^c \\ \lambda_{drn}^c \end{bmatrix} = \begin{bmatrix} -\frac{1}{\tau_s} & -\omega_{cn} & \frac{k_r}{\tau_s} & 0 \\ \omega_{cn} & -\frac{1}{\tau_s} & 0 & \frac{k_r}{\tau_s} \\ \frac{k_s}{\tau_r} & 0 & -\frac{1}{\tau_r} & (\omega_{rn} - \omega_{cn}) \\ 0 & \frac{k_s}{\tau_r} & (\omega_{cn} - \omega_{rn}) & -\frac{1}{\tau_r} \end{bmatrix} \begin{bmatrix} \lambda_{qsn}^c \\ \lambda_{dsn}^c \\ \lambda_{qrn}^c \\ \lambda_{drn}^c \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{qsn}^c \\ v_{dsn}^c \end{bmatrix} \quad (5.265)$$

where $k_s = L_m/L_s =$ stator coupling factor, $k_r = L_m/L_r =$ rotor coupling factor, $\sigma = 1 - L_m^2/(L_s L_r) = 1 - k_s k_r =$ leakage coefficient, $\tau_s = L_s/R_s =$ stator time constant, $\tau_r = L_r/R_r =$ rotor time constant, $\tau_s' = \sigma \tau_s =$ transient stator time constant, $\tau_r' = \sigma \tau_r =$ transient rotor time constant, $\tau = \omega_b t =$ normalized time. To gain further insight, various time constants and normalized time are introduced into this

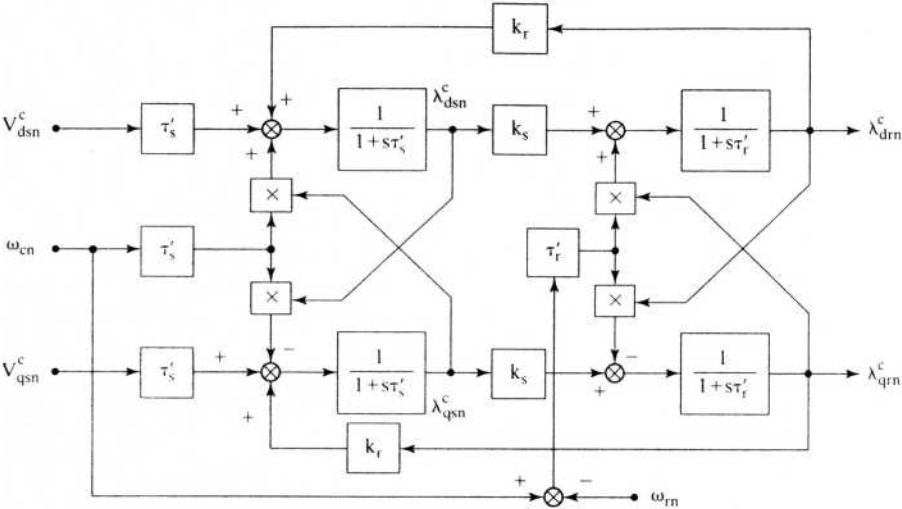


Figure 5.20 Signal-flow graph of the induction machine

model. The signal-flow graph of the system is shown in Figure 5.20. This does not give any additional advantage in analyzing the system and therefore is not considered any further.

The eigenvalues of the system as the rotor speed is increased from zero to 0.5 p.u. are shown in Figure 5.21 for a machine with normalized parameters of $R_{sn} = 0.0446$ p.u., $R_{rn} = 0.054$ p.u., $L_{mn} = 2.89$ p.u., $L_{sn} = 3.005$ p.u., and $L_{rn} = 3.13$ p.u. and in stator reference frames. The root loci on the left represent the two rotor eigenvalues; the other represents the stator eigenvalues. Their distinction comes from the fact that the rotor eigenvalues increase in their imaginary values as the rotor speed increases. As the stator does not experience the physical rotation, its eigenfrequencies are more or less stationary. The stator and rotor eigenvalues are complex conjugates; thus, it is sufficient to consider one of each to obtain a full picture of the dynamic characteristics of the machine. Note that by combining the two rotor and two stator equations into a resultant rotor and stator equation, the order of the system is reduced from four to two, resulting in the root loci containing one of the rotor and stator eigenvalues. That is precisely what the space-phasor model attempts to capture, with the attendant advantages, as shown in the succeeding sections. The real values denoting the transient time constants associated with the stator and rotor are to be noted; that they approach close to each other in value at higher speeds is also seen clearly.

5.11.3 Root Loci of the DQ Axes Based Induction Machine Model

Figure 5.22 shows the loci of the eigenvalues when the reference frames are fixed arbitrarily at a speed of 0.5 p.u. instead of at zero reference frames speed as in the Figure 5.21. The eigenfrequencies (the imaginary part of the eigenvalues) are shifted

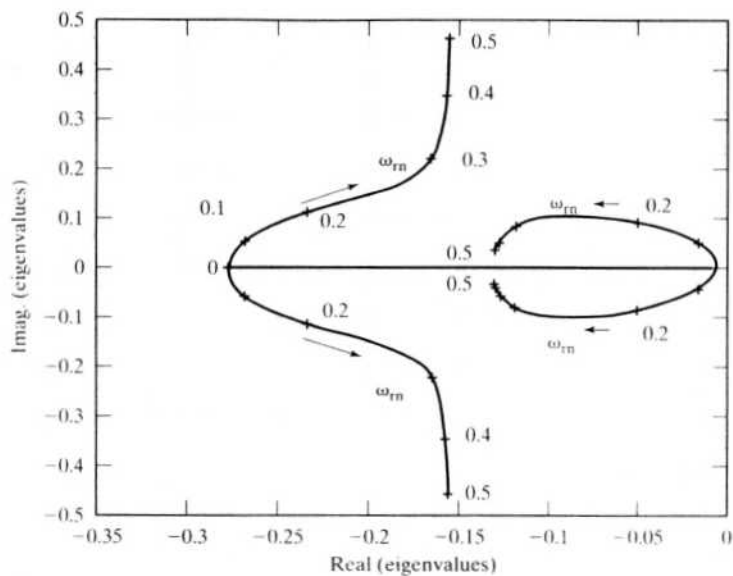


Figure 5.21 Loci of the eigenvalues in stator reference frames with rotor speed varying from 0 to 0.5 p.u.

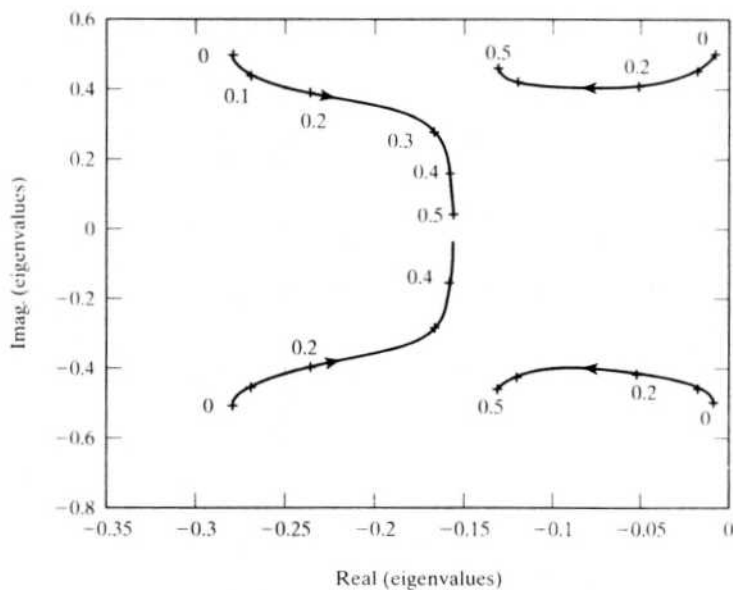


Figure 5.22 Loci of the eigenvalues in arbitrary reference frames with a velocity of 0.5 p.u. and rotor speed varying from 0 to 0.5 p.u.

by the reference-frames speed, i.e., 0.5 p.u. in both the stator and rotor root loci. The negative eigenfrequencies are added to the reference-frames speed, whereas the positive eigenfrequencies are subtracted from the reference-frames speed, as becomes clearly discernible in this figure. In effect, the root loci computed for one frame are convertible to another reference frame by simply adding and subtracting the reference-frames speed from their eigenfrequencies. This has great significance in that it demonstrates the interrelationship of the machine eigenvalues in various reference frames.

Consider that every eigenvalue contributes to a response term of exponential nature, with one, due to the real part, indicating the damping and the other, due to the eigenfrequency term (i.e., imaginary value of the root), in the form of a phase. The reference frames merely displace the eigenfrequencies so that the phase term of the response cancels the phase rotation added to the input variables, resulting in the elimination of the contribution from the reference-frames speed term in the responses, but by having intact the real parts regardless of the reference frames, the damping is preserved uniformly in the responses in all of the reference frames.

An alternative approach is to look at the induction machine with a set of two windings, one in the stator and the other on the rotor, using a space-phasor approach.

5.11.4 Space-Phasor Model Derivation

Let the space phasor of the stator flux linkages in arbitrary reference frames be given by

$$\lambda_{sn}^c = \lambda_{qsn}^c - j\lambda_{dsn}^c \quad (5.266)$$

This definition is valid for currents, voltages, and flux linkages in the machine for both the stator and rotor variables. Such a relationship can alternatively be expressed in complex phasor form as

$$\lambda_{sn}^c = e^{-j\omega_{cn}t} \left\{ \frac{2}{3} (\lambda_{asn} + e^{j2\pi/3} \lambda_{bsn} + e^{j4\pi/3} \lambda_{csn}) \right\} \quad (5.267)$$

where the first exponential term gives the reference-frame rotation at an arbitrary speed of ω_{cn} . Note that the operation in the parentheses denotes the resolution of the three phases from abc axes to the q and d axes, with the q axis aligned to phase A. The resolution from abc to qdo axes is achieved by projecting the abc axes variables onto the q and d axes through simple trigonometric relationships developed in section 5.7.3. Further, the expression inside the parentheses, on expansion, results in

$$\begin{aligned} \frac{2}{3} (\lambda_{asn} + e^{j2\pi/3} \lambda_{bsn} + e^{j4\pi/3} \lambda_{csn}) &= \frac{2}{3} \left\{ \lambda_{asn} - 0.5(\lambda_{bsn} + \lambda_{csn}) - j\frac{\sqrt{3}}{2} (\lambda_{csn} - \lambda_{bsn}) \right\} \\ &= \left\{ \lambda_{asn} - j\frac{1}{\sqrt{3}} (\lambda_{csn} - \lambda_{bsn}) \right\} = \lambda_{qsn} - j\lambda_{dsn} \quad (5.268) \end{aligned}$$

The vector rotation at the reference-frame speed converts these stator reference-frame variables into the arbitrary reference-frame variables. Similarly, to recover the phase variables in stator reference frames from the arbitrary reference-frame variables, the procedure is reversed. First, the variables are separated into the real and imaginary values, which are then identified as the q and d axes variables in the arbitrary reference frames. Using the qdo -to- abc axes transformation T_{abc}^{-1} (developed in section 5.7.3), the variables in the stator reference frames are recovered. Alternately, it could also be derived compactly as

$$\begin{aligned}\lambda_{asn} &= \operatorname{Re} \left\{ e^{j\omega_{cn}t} \lambda_{sn}^c \right\} \\ \lambda_{bsn} &= \operatorname{Re} \left\{ e^{j\omega_{cn}t} e^{j\frac{4\pi}{3}} \lambda_{sn}^c \right\} \\ \lambda_{csn} &= \operatorname{Re} \left\{ e^{j\omega_{cn}t} e^{j\frac{2\pi}{3}} \lambda_{sn}^c \right\}\end{aligned}\quad (5.269)$$

Applying the space-phasor definition given in the above to the expressions given in the matrix equation (5.265), the following system of equations is derived in arbitrary reference frames:

$$\begin{aligned}\frac{d\lambda_{sn}^c}{d\tau} + \left(\frac{1}{\tau_s'} + j\omega_{cn} \right) \lambda_{sn}^c &= \frac{k_r}{\tau_s'} \lambda_{rn}^c + v_{sn}^c \\ \frac{d\lambda_{rn}^c}{d\tau} + \left[\frac{1}{\tau_r'} + j(\omega_{cn} - \omega_{rn}) \right] \lambda_{rn}^c &= \frac{k_s}{\tau_r'} \lambda_{sn}^c\end{aligned}\quad (5.270)$$

where the voltage and current phasors are similarly defined. The signal-flow graph for these equations is shown in Figure 5.23. It is very compact and lends itself elegantly to a similarity with the dc machine, with which the engineers are much more familiar than they are with the multiphase ac machines. It consists of two time constants, corresponding to the armature and field of the dc machine, but the windings are coupled in the induction machine, unlike in the dc

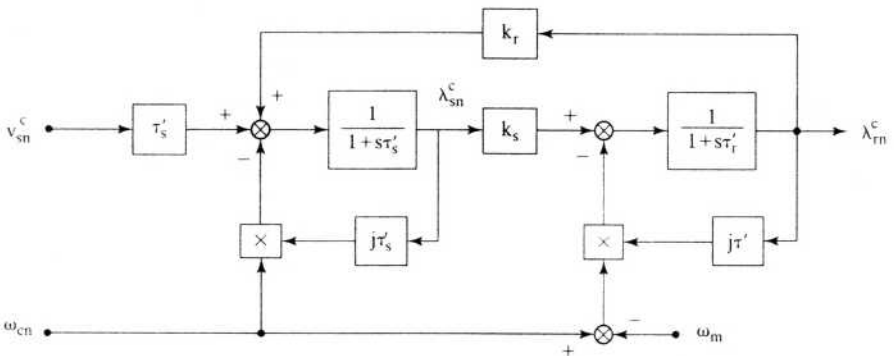


Figure 5.23 Signal-flow graph of the space-phasor-modeled induction machine

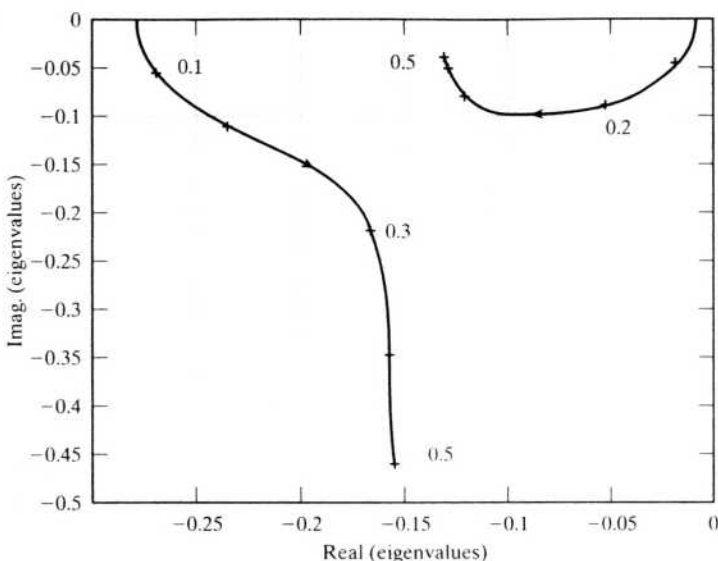


Figure 5.24 Loci of the eigenvalues in stator reference frames, with the rotor speed varied from 0 to 0.5 p.u. in a space-phasor model

machine. In the dc machine, a coupling exists only between the field and armature but not vice versa. The space-phasor modeling is uniquely suitable only for induction machines, because of its symmetry between the stator and the rotor. Such symmetry does not exist in the permanent-magnet synchronous machines, where the direct and quadrature axis inductances are not equal, because of the saliency of the rotor, and therefore the space-phasor model is not attempted in that case.

5.11.5 Root Loci of the Space-Phasor Induction-Machine Model

The root loci of the space-phasor-model-based induction machine for the same constants used in the previous figures are shown in the stator reference frames in Figure 5.24. Note that there exist only two eigenvalues in this system, and they correspond to one stator and rotor eigenvalue from the qd model with their complex conjugates removed. However, the elegance of this model comes into play in the compactness of its expression and the insight gained through it in comparison to the qd model. The eigenvalues for any arbitrary reference frames are obtained by adding the reference-frames speed to the eigenfrequencies of the roots of the stator reference frames, much like the qd model-based eigenvalues.

5.11.6 Expression for Electromagnetic Torque

The electromagnetic torque in S.I. units is found to be

$$T_e = \frac{3}{2} \frac{P}{2} \frac{L_m}{\Delta_l} \{ \lambda_{qs}^c \lambda_{dr}^c - \lambda_{ds}^c \lambda_{qr}^c \} \quad (5.271)$$

and in normalized units is derived as

$$T_{en} = \frac{T_e}{T_b} = \frac{T_e}{\left(\frac{3}{2} \frac{P}{2} \frac{V_b I_b}{\omega_b} \right)} \quad (5.272)$$

Note that

$$\begin{aligned} V_b &= \omega_b \lambda_b \\ \lambda_b &= L_b I_b \end{aligned} \quad (5.273)$$

where L_b is the base inductance. Substituting the above relationships into the torque expression results in

$$T_{en} = \frac{T_e}{\left(\frac{3}{2} \frac{P}{2} \frac{\lambda_b^2}{L_b} \right)} \quad (5.274)$$

which, on substitution of T_e , leads to the normalized torque as

$$T_{en} = \frac{L_{mn}}{(L_{sn} L_{rn} - L_{mn}^2)} \{ \lambda_{qsn}^c \lambda_{drn}^c - \lambda_{dsn}^c \lambda_{qrn}^c \} \quad (5.275)$$

The normalized torque can be written compactly also in terms of the stator and rotor flux linkage phasors as

$$T_{en} = \frac{L_{mn}}{(L_{sn} L_{rn} - L_{mn}^2)} \{ \text{Im}(\lambda_{sn}^c \overline{\lambda_{rn}^c}) \} = \frac{L_{mn}}{\sigma L_{sn} L_{rn}} \{ \text{Im}(\lambda_{sn}^c \overline{\lambda_{rn}^c}) \} \quad (5.276)$$

where $\overline{\lambda_{rn}^c}$ is the complex conjugate of the rotor flux linkage phasor and Im implies the imaginary component of the variable inside the parentheses. This lends itself to the interpretation that the stator and rotor flux linkage phasors are interacting to provide an electromagnetic torque, very much as the armature and field flux linkages in a dc machine interact to produce the electromagnetic torque. Similarly, the normalized torque could also be written as

$$T_{en} = (i_{qsn}^c \lambda_{dsn}^c - i_{dsn}^c \lambda_{qsn}^c) = \text{Im}(i_{sn}^c \overline{\lambda_{rn}^c}) \quad (5.277)$$

This expression portrays the torque as the interaction of the stator current and stator flux linkage phasors; similarly, the following expression gives it in terms of the interaction between the stator current and rotor flux linkage phasor:

$$T_{en} = \frac{L_m}{L_r} \{ \text{Im}(i_{sn}^c \overline{\lambda_{rn}^c}) \} \quad (5.278)$$

Various torque expressions contribute to a variety of control schemes for induction motors, as demonstrated in Chapter 8 on vector-controlled induction-motor drives.

Example 5.8

The parameters of an induction motor are

2000 hp, 2300 V, 4 pole, 60 Hz, star-connected stator

$$R_s = 0.02 \, \Omega; R_r = 0.12 \, \Omega; L_m = 0.1326 \, \text{H}; L_s = L_r = 0.1335 \, \text{H}$$

$$J = 10 \, \text{kg-m}^2; B = 0; \text{Load torque is zero.}$$

Plot (i) the free-acceleration characteristics of the machine and (ii) the dynamic performance when the machine is started with linearly varying input voltages from 0.03 p.u. to 1 p.u. in one second. The space-phasor model of the machine in stator reference frames is recommended for solving this problem.

Solution (i) The stator reference-frames model of the induction machine in SI units derived in section 5.7.7.1 is considered here for deriving the space-phasor model. The four dq equations in (5.128) are reduced to two space-phasor equations by the following steps:

$$\vec{v}_s = v_{qs} - jv_{ds} = (R_s + L_s p)(i_{qs} - ji_{ds}) + L_m p(i_{qr} - ji_{dr}) = (R_s + L_s p)i_s + L_m p i_r$$

Similarly, the rotor equations are combined as

$$v_r = v_{qr} - jv_{dr} = (R_r + L_r p - j\omega_r L_r)i_r + L_m(p - j\omega_r)i_s$$

Then these two equations are cast in state-space form as

$$\begin{bmatrix} \dot{p}i_s \\ \dot{p}i_r \end{bmatrix} = \frac{1}{(L_s L_r - L_m^2)} \begin{bmatrix} (-L_r R_s + jL_m^2 \omega_r) & L_m(R_r - j\omega_r L_r) \\ L_m(R_s + j\omega_r L_s) & L_s(-R_r + j\omega_r L_r) \end{bmatrix} \begin{bmatrix} p i_s \\ p i_r \end{bmatrix} + \begin{bmatrix} L_r \\ -L_m \end{bmatrix} v_s$$

The air gap torque is derived as

$$T_e = \frac{3}{2} \frac{P}{2} L_m (i_{qs} i_{dr} - i_{ds} i_{qr}) = \frac{3}{2} \frac{P}{2} L_m [\text{Imag}(i_s \bar{i}_r)]$$

The mechanical equation is given by

$$J \frac{d\omega_r}{dt} = \frac{P}{2} T_e$$

because the friction and load torques are zero for the given problem. The three differential equations are solved by numerical integration, and the phasor magnitudes of stator voltage, current and flux linkages and rotor flux linkages, air gap torque, and rotor speed are plotted in normalized units, as shown in Figure 5.25. The stator and rotor flux linkages phasors are derived from the stator and rotor currents as

$$\lambda_s = \lambda_{qs} - j\lambda_{ds} = (L_s i_{qs} + L_m i_{qr}) - j(L_s i_{ds} + L_m i_{dr}) = L_s i_s + L_m i_r$$

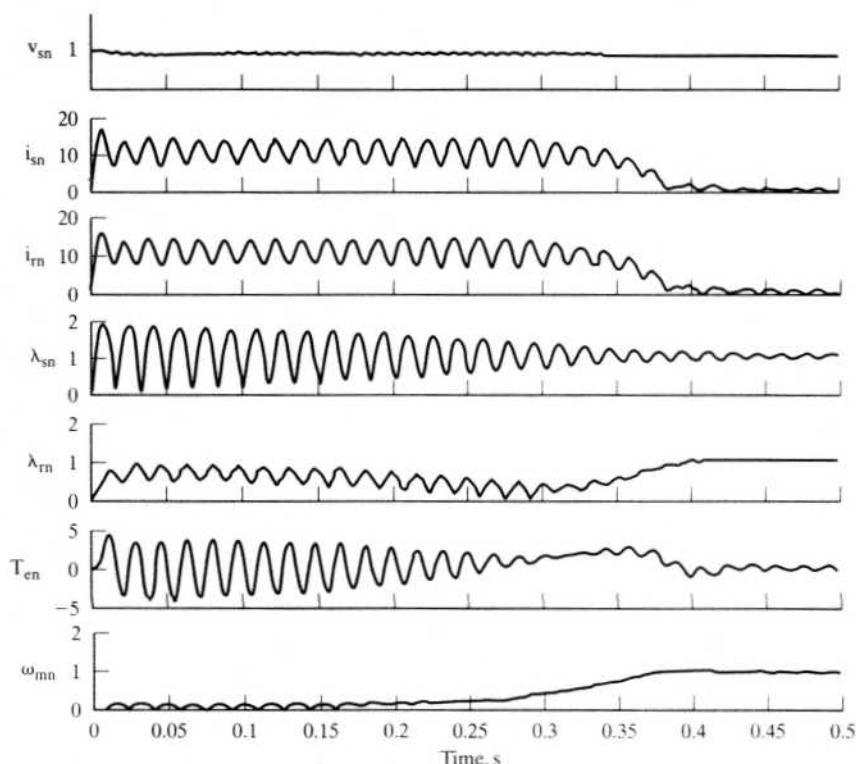


Figure 5.25 Line-start performance of an induction machine in space-phasor variables

Similarly, the rotor flux-linkages phasor is derived as

$$\lambda_r = \lambda_{qr} - j\lambda_{dr} = (L_r i_{qr} + L_m i_{qs}) - j(L_r i_{dr} + L_m i_{ds}) = L_r i_r + L_m i_s$$

Note that only phasor magnitudes are shown in the simulation results. The space-phasor simulation has reduced the total number of differential equations from 5 to 3, but the reduction comes at the expense of their being in complex variables. Even though there is no distinct advantage in simulating the system in complex variables, significant insight can be gained by viewing the phasors rather than dq components of the key variables.

(ii) Linearly varying the input voltages has limited the stator currents to 7 p.u., as shown in Figure 5.26. It has eliminated oscillations in the currents and flux linkages and hence has eliminated entirely the torque pulsations. The start-up time has increased by 75% as compared to the line start with full supply voltages. Even in this method, multiple times the rated resistive losses occur, and hence the starts per hour are restricted for large machines such as the one simulated here.

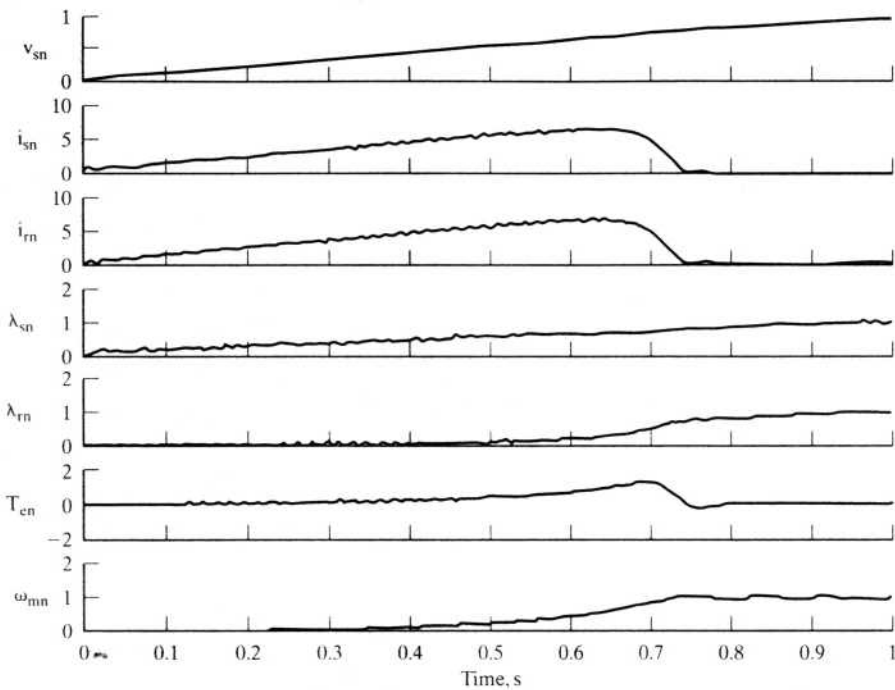


Figure 5.26 Start-up performance of an induction machine with linear variation of input-stator voltage phasor

5.11.7 Analytical Solution of Machine Dynamics

The space-vector approach not only provides compact and conceptual representation of the induction machine but also assists in the direct closed-form solution of the dynamic processes in the machine, neglecting mechanical dynamics. The solution of the stator and rotor flux-linkage phasors for a step input in the stator voltage phasor can be analytically found; it involves the solution of only two first-order differential equations with complex coefficients. This, in detail, is given below.

The roots of the system, p_1 and p_2 , are found from the space-vector model derived from equation (5.272) as

$$\dot{x} = Ax + Bu$$

where

$$\begin{aligned} A &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}; B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; x = [\lambda_{sn}^c \quad \lambda_{rn}^c]; u = \begin{bmatrix} v_{sn}^c \\ 0 \end{bmatrix} \\ a_{11} &= -\left(\frac{1}{\tau_s} + j\omega_{cn}\right); a_{12} = \frac{k_r}{\tau_s}; a_{21} = \frac{k_s}{\tau_r}; a_{22} = -\left\{\frac{1}{\tau_r} + j(\omega_{cn} - \omega_{rn})\right\} \end{aligned} \tag{5.279}$$

where P is the number of poles.

The differential speed between the stator magnetic field and rotor windings is slip speed, and that is responsible for the frequency of the induced emfs in the rotor and hence their currents. Therefore, the rotor currents are at slip frequency, which can be obtained from the angular slip speed by dividing it by 2π . The slip is defined as

$$s = \frac{\omega_{sl}}{\omega_s} \quad (5.7)$$

Note that slip is nondimensional. It is proven later that this is one of the most important variables in the control and operation of the induction machines. Combining equations (5.6) and (5.7), the rotor electrical speed is given as

$$\omega_r = \omega_s(1 - s), \text{ rad/sec} \quad (5.8)$$

From this, the rotor speed in rpm, denoted by n_r , is expressed as

$$n_r = n_s(1 - s)(\text{rpm}) \quad (5.9)$$

where n_s is the synchronous speed or the speed of the stator magnetic field in rpm, given by

$$n_s = \frac{120f_s}{P} \text{ rpm} \quad (5.10)$$

5.3 INDUCTION-MOTOR EQUIVALENT CIRCUIT

The equivalent circuit of the induction motor is very similar to that for a transformer. Although the rotor currents are at slip frequency, the rotor is incorporated into the circuit in a simple way. Recognizing the fact that the stator and rotor windings have resistances and leakage inductances and that the mutual inductance for modeling the mutual flux links the stator and rotor windings, Figure 5.1 gives the elementary equivalent circuit. The parameters are stator resistance per phase, R_s , rotor resistance per phase, R_{rr} , mutual inductance, L_m , stator leakage inductance

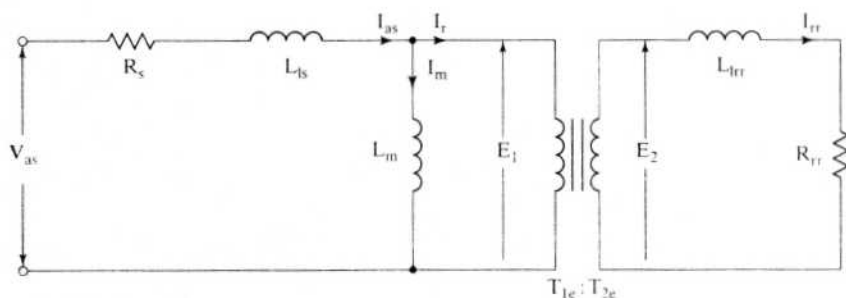


Figure 5.1 Elementary equivalent circuit for the induction motor

The roots of the system are found from the characteristic equation, i.e., $\det [A - pI] = 0$. The roots, p_1 and p_2 , are evaluated as

$$p_1, p_2 = \frac{1}{2} \{a \pm \sqrt{a^2 - 4b}\}$$

where

$$a = a_{11} + a_{22}; b = a_{11}a_{22} - a_{12}a_{21} \quad (5.280)$$

In taking the square root of the discriminant, which is a complex number, use has to be made of DeMoivre's theorem converted into exponential form. After having solved for the roots, and keeping the stator frequency and rotor angular speed constant, you can derive the stator and rotor flux linkage phasors for a unit step input of the stator voltage phasor of $-j1$ p.u. as

$$\begin{aligned} \lambda_{sn}^c(\tau) &= \frac{1}{(p_2 - p_1)} \left[\left\{ -(p_1 + a_{22}) \left(\lambda_{sn}^c(0) + \frac{j}{p_1} \right) \right. \right. \\ &\quad \left. \left. + a_{12} \lambda_{rn}^c(0) \right\} e^{-p_1 \tau} + \left\{ (p_2 + a_{22}) \left(\lambda_{sn}^c(0) + \frac{j}{p_2} \right) - a_{12} \lambda_{rn}^c(0) \right\} e^{-p_2 \tau} \right] + j \frac{a_{22}}{p_1 p_2} \\ \lambda_{rn}^c(\tau) &= \frac{1}{(p_2 - p_1)} \left[\left\{ a_{21} \left(\frac{j}{p_1} + \lambda_{sn}^c(0) \right) - (p_1 + a_{11}) \lambda_{rn}^c(0) \right\} e^{-p_1 \tau} \right. \\ &\quad \left. + \left\{ -a_{21} \left(\frac{j}{p_2} + \lambda_{sn}^c(0) \right) + (p_2 + a_{11}) \lambda_{rn}^c(0) \right\} e^{-p_2 \tau} \right] - j \frac{a_{21}}{p_1 p_2} \end{aligned} \quad (5.281)$$

where $\lambda_{sn}^c(0)$ and $\lambda_{rn}^c(0)$ are the initial values of the stator and rotor flux linkage phasors, respectively. Note that the normalized time is used here. The rotor and stator flux-linkage phasors are spatial distributions of the flux linkages and are available as a function of motor parameters and initial conditions. These analytical solutions are easily obtained compared to the dq axes model case. They help in understanding the magnetic-energy propagation within the machine during transients [14, 17].

5.11.8 Signal-Flow Graph of the Space-Phasor-Modeled Induction Motor

Based on the system of equations derived in the above, the signal-flow graph of the induction machine is drawn as shown in Figure 5.23. The symbol s denotes the Laplace operator in this figure. Addition of the mechanical load to the machine and load inertia, friction constant, and load torque will complete the block-diagram representation of the induction motor. In this case, the system involves the solution of a third-order system in complex variables. From this point onward, the space-phasor modeling does not provide an advantage over the dq model: numerical solution is resorted to in the solution of linearized systems of equations of third order in the case of space-phasor approach, as against the fifth order in the dq model, and the elegance of the meaningful analytical solution is lost in both cases.

Example 5.9

Consider a set of three-phase sinusoidal input voltages to an induction motor with a frequency of ω_s . Express the stator-voltage phasor in arbitrary reference frames, and, hence, find from it the stator-voltage phasors in the rotor and synchronous reference frames.

Solution The three-phase sinusoidal input voltages are defined to be

$$v_{as} = V_m \sin \theta_s; v_{bs} = V_m \sin \left(\theta_s - \frac{2\pi}{3} \right); v_{cs} = V_m \sin \left(\theta_s + \frac{2\pi}{3} \right)$$

The stator-voltage phasor in stator reference frames is given by

$$\mathbf{v}_s = \frac{2}{3} (v_{as} + e^{j\frac{2\pi}{3}} v_{bs} + e^{j\frac{4\pi}{3}} v_{cs})$$

Substituting for the phase voltages from the definitions and expanding them with the exponential terms simplifies the stator-voltage phasor to

$$\mathbf{v}_s = V_m (\sin \theta_s - j \cos \theta_s)$$

The stator-voltage phasor in the arbitrary reference frame is given by

$$\mathbf{v}_s^c = \mathbf{v}_s e^{-j\theta_c} = \mathbf{v}_s e^{-j\theta_s}$$

By substituting for the voltage phasor in stator reference frames into the above equation, the stator-voltage phasor in arbitrary reference frames is derived as

$$\mathbf{v}_s^c = V_m (\sin \theta_s - \cos \theta_s) e^{-j\theta_c} = V_m \{-j e^{j\theta_c}\} e^{-j\theta_c} = -j V_m e^{j(\theta_s - \theta_c)}$$

from which the stator-voltage phasors in rotor reference frames and synchronous frames are found by substituting $\theta_c = \theta_r$ and $\theta_c = \theta_s$, respectively, in the stator-voltage phasor expression in the arbitrary reference frames:

$$\begin{aligned} \mathbf{v}_s^r &= V_m \{\sin(\theta_s - \theta_r) - j \cos(\theta_s - \theta_r)\} \\ \mathbf{v}_s^e &= V_m \{0 - j1\} \end{aligned}$$

5.12 CONTROL PRINCIPLE OF THE INDUCTION MOTOR

The control principle of the induction machine is derived in this section. A physical interpretation is given that indicates its close resemblance to the separately-excited dc machine. A transient-free operation of the induction machine is achieved only if the stator (or for that matter rotor) flux-linkages phasor is maintained constant in its magnitude and its phase is stationary with respect to the current phasor (or the current phasor is varied in such a way that it counterbalances the rate of change of the flux-linkages phasor). This fundamental theorem for the control of ac machines is proved in the following for induction machines. Such an understanding is crucial to developing appropriate control techniques and their

implementations, including vector-controlled drives of various kinds, with and without position and speed sensors.

Consider the induction-machine model in the synchronous reference frame. In this frame, the sinusoidal variables become dc quantities, i.e., constants, and hence their derivatives, such as those of the currents and flux linkages in steady state, are all zero. This step helps in visualizing the steady state much more simply than the models in other reference frames. The phasor is defined to be the resultant of the respective q and d axes variables such as currents, voltages, and flux linkages. The stator current phasor and stator flux-linkages phasor are defined as

$$\begin{aligned} i_s &= i_{qs}^e - j i_{ds}^e \\ \lambda_s &= \lambda_{qs}^e - j \lambda_{ds}^e \end{aligned} \quad (5.282)$$

The input power for balanced supply voltages is given by

$$p_i = \frac{3}{2} (v_{qs}^e i_{qs}^e + v_{ds}^e i_{ds}^e) \quad (5.283)$$

Substituting for the voltages in terms of the machine parameters, currents, and stator angular frequency, the input power is obtained as

$$p_i = \frac{3}{2} (R_s [(i_{qs}^e)^2 + (i_{ds}^e)^2] + \omega_s L_m [i_{qs}^e i_{dr}^e - i_{ds}^e i_{qr}^e] + i_{qs}^e [L_s p i_{qs}^e + L_m p i_{qr}^e] + i_{ds}^e [L_s p i_{ds}^e + L_m p i_{dr}^e]) \quad (5.284)$$

where p is the differential operator and ω_s is the stator frequency at which the reference frames are rotating. Recognizing the electromagnetic torque as

$$T_e = \frac{3}{2} \frac{P}{2} L_m (i_{qs}^e i_{dr}^e - i_{ds}^e i_{qr}^e) = \frac{3}{2} \frac{P}{2} (\lambda_{ds}^e i_{qs}^e - \lambda_{qs}^e i_{ds}^e) \quad (5.285)$$

where the stator d and q axes flux linkages are given by

$$\begin{aligned} \lambda_{qs}^e &= L_s i_{qs}^e + L_m i_{qr}^e \\ \lambda_{ds}^e &= L_s i_{ds}^e + L_m i_{dr}^e \end{aligned} \quad (5.286)$$

gives the power input in terms of the torque and flux linkages and currents, by substituting equations (5.285) and (5.286) into equation (5.284), as

$$p_i = \frac{3}{2} R_s [(i_{qs}^e)^2 + (i_{ds}^e)^2] + \frac{2}{P} \omega_s T_e + \frac{3}{2} [i_{qs}^e p \lambda_{qs}^e + i_{ds}^e p \lambda_{ds}^e] \quad (5.287)$$

By substituting the phasor in the place of the individual q and d components of the currents and stator flux linkages, and by substituting for the electromagnetic torque in terms of the stator current and stator flux-linkages phasors, the input power is compactly written as

$$p_i = \frac{3}{2} R_s i_s^2 + \frac{2}{P} \omega_s \text{Im}[i_s \bar{\lambda}_s^e] + \frac{3}{2} \text{Re}[i_s p \bar{\lambda}_s^e] \quad (5.288)$$

where Re and Im are the real and imaginary part of the functions, respectively, and $\bar{\lambda}_s$ is the complex conjugate of the stator flux-linkages phasor. There are three distinct components of the input power, as is seen from equation (5.288). The first term corresponds to the stator resistance losses; the remaining two terms constitute the power crossing the stator and entering into the air gap. The second term corresponds to the sum of the rotor slip power and mechanical power, because the stator frequency is the sum of the slip and rotational frequencies. The latter part of the second term supplies shaft power and friction and windage losses in the machine. The slip power corresponds to the rotor resistance losses. Note that the electromagnetic torque is the product of the stator current and flux-linkages phasors, as in the case of the dc machine, where it is equal to the product of the field flux linkages and armature current. This expression makes the direct equivalence between the dc and induction machines complete. The understanding that came out of this is that the induction machines could also have equivalent control, i.e., decoupling control, like that of the dc machines. The only difference in the induction machine is that the variables are phasors, unlike in the dc machines, where they are scalars. In the case of the induction machine, the flux linkages have to be controlled through the stator-current phasor only, which also controls the electromagnetic torque. Thereby, only one variable, the current phasor alone, controls both torque and flux-producing channel currents in the machine, resulting in an inherent coupling of these channels. By knowing the instantaneous position of the stator flux linkages and current phasors, they can be controlled to occupy any position, with the desired magnitudes resulting in a class of control schemes known as vector control.

The third term in the input-power expression denotes the rate of change of magnetic energy. In steady state, this term is zero: the stator flux-linkages phasor is a constant, hence its derivative is zero; but this term need not be zero during transients. Only if it is made zero at all times does the power drawn by the machine go only to provide for the stator resistance losses and synchronous power to meet the load demands and rotor copper losses. This results in minimum exchange of power between the power supply and the induction machine, enabling the transient free operation of the induction machine even during dynamic load or speed changes. Such an operation is desirable when the induction machine is supplied from a finite source, such as an inverter, whose capacity is very limited. Then the key to the control of the induction machine is that this term has to be identically zero at all times of its operation. This term can be made zero only under one of the following conditions:

- (i) The rate of change of the stator flux-linkages phasor is zero. This is possible only when the stator flux-linkages phasor remains constant in magnitude when seen from the synchronous reference frames. Also, for the derivative to be zero, the phase of the stator flux linkages is constant in synchronous reference frames. In stator reference frames, the phase of the stator flux-linkages phasor has to be synchronous with the current phasor, i.e., the differential velocity between them has to be zero. Therefore, the phase angle between the stator current phasor and stator flux-linkages phasor, known as

torque angle, has to be a constant when viewed from both the stator and synchronous reference frames. Note that keeping the derivative of the stator-flux linkages phasor zero is not always possible; sometimes, in the case of flux weakening to increase the range of speed, it may be intentionally varied from its rated value. Then the change of the stator flux-linkages phasor is not zero, leading to a transient operation in the machine.

- (ii) The real part of the product of the stator current phasor and derivative of the stator flux-linkages phasor is zero. This condition is required to be satisfied for transient-free operation during intentional flux-linkages variation in the machine. This is achieved by adjusting the current phasor in such a way that the resulting flux-linkages variation is obtained while the real part of the product of the stator current and derivative of the flux-linkages phasors becomes identically zero.
- (iii) The stator current phasor is zero; this is a trivial condition and hence is neglected in further consideration.

Note that, to achieve condition (i) or (ii), the stator flux-linkages phasor has to be known in both its magnitude and its position, along with the current phasor. From this information, the relative position between them can be used to control both the electromagnetic torque and stator flux linkages, resulting in a transient-free operation of the induction machine considered in detail in Chapter 8.

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5.14 DISCUSSION QUESTIONS

1. Is it possible to operate the induction machine as a generator? What is the polarity of slip in that mode of operation?
2. Cite instances in which the machine may be used to regenerate.
3. The steady-state model of the induction machine is generally assumed to have constant parameters. It is known that all those parameters vary with operating conditions. Discuss the method by which parameter variations can be considered in the computation of steady-state performance.
4. Besides core and copper losses, what other losses need to be considered in the performance evaluation of the induction motors?
5. Prove that the relative speed between stator and rotor mmfs is zero. What is the primary consequence if their speeds are unequal?
6. Discuss the shortcomings of the induction-motor dynamic model from the viewpoint of losses.
7. In this computer age, why not use the actual motor model with time-varying inductances instead of resorting to d and q axes models?
8. Discuss the suitability of stationary reference frames for the induction motor in power-system studies.
9. Can the d and q models be used for supply-voltage-unbalance studies?
10. If the input voltages/currents are rectangular or trapezoidal in wave shapes, explain how the d and q models are used to predict the steady-state behavior. (Hint: Any periodic waveform can be resolved into a fundamental and a number of harmonics with sinusoidal wave shapes.)

11. Will the type (Y or Δ) of the stator connection change the dynamic models that have been derived in this chapter?
12. Will the parameter sensitivity affect the stability of the induction motor? (Hint: Reference 11.)
13. Which is the most important transfer function for the induction motor? Why? (Hint: Reference 13.)
14. The induction motor responds differently to various inputs during dynamic situations. Discuss the merits of one input over another and how to quantify their effects. (Hint: Reference 10.)
15. The machine parameters could have wide variations due to continued operation, with resultant increase in temperature due to losses, changes in input voltages that vary the magnetic saturation, and frequency variation. Discuss the significance and magnitude of variation in parameters due to these changes.
16. The modeling does not account for stray losses. These losses are difficult to predict and usually vary from 0.5 to 1% of the input power. Discuss ways of including the stray losses in performance calculations and their significance in prediction.
17. The friction and windage losses are not included in the steady-state equivalent circuit. Could they be included in the form of resistive losses similar to the core losses?
18. How could magnetic saturation be included in the steady-state and transient modeling of induction motors?
19. Is it possible to operate the induction motor in the statically unstable region?
20. During free acceleration, the electromagnetic torque has large oscillations. What are the consequences of such oscillations?
21. Is it possible to eliminate or mitigate these oscillations with a constant-frequency ac supply?
22. With the availability of variable-voltage and variable-frequency sources, could they be used for the damping of the oscillations?

5.15 EXERCISE PROBLEMS

1. The slip speed at breakdown can be approximately represented in terms of the rotor and stator parameters. Derive this expression.
2. An induction motor has the following ratings and parameters:

40 hp, 460 V, 4 pole, 3 phase, 60 Hz, Y connected

$R_s = 0.22 \, \Omega$, $R_r = 0.209 \, \Omega$, $L_m = 0.04 \, \text{H}$

$L_s = 0.0425 \, \text{H}$, $L_r = 0.043 \, \text{H}$, $B = 0$

Load torque = $T_l = 0$, $J = 0.124 \, \text{kg} \cdot \text{m}^2$, $a = 2$

The statically stable slip region is required to be doubled. That can be achieved by connecting external resistors in the rotor phases. Calculate approximately the value of external rotor resistance/phase to be added.

3. Neglecting motor copper losses and core losses, prove that the efficiency of the induction motor is equal to $(1 - s)$. Discuss the implications of this derivation over the variable-speed operation of the induction motor when the applied motor voltages are varied. (More will be given in Chapter 6 on this.)

4. Capacitors are connected to the stator windings of the induction motor. Derive an expression to quantify the improvement in power factor due to the capacitors.
5. The starting torque can be increased by increasing rotor resistances with the insertion of external resistances in slip-ring induction rotor motors. Find the external resistor per phase to be introduced, in terms of the stator and rotor resistances and total leakage inductances, to double the starting torque. Consider an equivalent circuit, neglecting the magnetizing branch.
6. Consider a current source at 60 Hz supplying the induction motor given in problem 2 with rated stator currents. Draw its torque-vs.-slip characteristics and compare them with the characteristics obtained by using a voltage source at rated voltages and 60 Hz. Comment on the salient features. (More will be given in Chapter 7 on this.)
7. A NEMA D motor with the following parameters is chosen to drive a pump. The load-torque characteristic of the pump is modeled as simply proportional to the square of the rotor speed and at rated speed requires rated torque for its operation. Its speed has to be varied from rated value to the lowest value, 0.7 p.u. Compute the efficiencies at 0.7 p.u. speed when the speed is varied by using rotor resistance control and alternatively by varying the stator input voltages. The parameters and ratings of the motors are

100 hp, 460 V, 4 poles, 3 ph, 60 Hz, star connected, 1050 rpm, $R_s = 0.05 \Omega$,
 $R_r = 0.2885 \Omega$, $L_m = 0.01856 \text{ H}$, $L_r = L_s = 0.01936 \text{ H}$.

8. Consider the motor given in Example 5.6, running in steady state with rated torque and supplied from a 200 V, 60 Hz, 3-phase ac main. It is desired to stop the motor quickly by cutting off supply to the stator windings and by shorting them. This is known as plugging. Analyze this operation, using dynamic simulation in the stator reference frames.
9. An induction motor is operating at rated values in steady state. The ac power supply is shut off for 0.1 second and reinstated at the end of 0.1 second. Analyze this situation from dynamic simulation results. The parameters and ratings of the motors are

1.5 MW, 1800 V(line to line), 566 A, 1480 rpm, 50 Hz, star connected, $R_s = 0.01934 \Omega$,
 $R_r = 0.01343 \Omega$, $L_m = 0.0305 \text{ H}$, $L_r = L_s = 0.0315 \text{ H}$, $B = 0.02 \text{ N-m/rad/sec}$, $J = 30 \text{ kg-m}^2$.

(Hint: This is known as reclosing transients in the induction motor. It is instructive to look at the rotor fluxes during the off-time and the induced emfs in the stator windings to appreciate the phase shift between the incoming supply voltages and induced emfs.)

10. Assume an induction motor is running in steady state with rated input voltages and full-load torque. Analyze the situation when two of the three phases are interchanged by

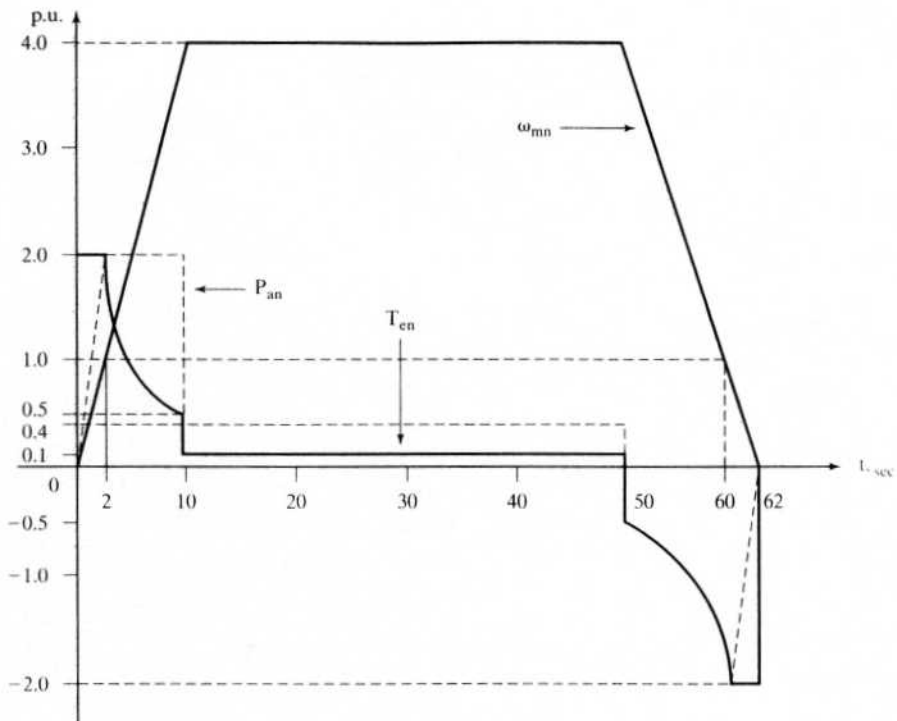


Figure 5.27 Load duty cycle for the motor

using dynamic simulation. This results in change of phase sequence to the machine. Comment on the results. The induction motor given in Example 5.6 can be used for the simulation.

11. An induction motor is desired for an application with the load duty cycle requiring the speed and air gap power shown in Figure 5.27. The absolute air gap power is maintained at 2 p.u. whenever the speed is above 1 p.u., and the load torque is only 0.1 p.u. at 4 p.u. speed. Calculate the nominal torque of the induction motor to be chosen for this application. (Hint: The rms value of the electromagnetic torque is the nominal torque.)

per phase, L_{ls} , rotor leakage inductance per phase, L_{lrr} , stator turns per phase, T_1 , rotor turns per phase, T_2 , induced emf in the stator per phase, E_1 , and induced emf in the rotor per phase, E_2 . Effective stator or rotor turns per phase is equal to the product of the number of turns per phase and the winding factor of the stator and rotor and, respectively, is denoted as k_{w1} and k_{w2} . The winding factor is the product of pitch and distribution factors that account for the specific winding characteristics. The evaluation of these factors can be found in any standard textbook on electrical machines.

The relationship between the induced emfs is

$$\frac{E_2}{E_1} = s \frac{T_{2e}}{T_{1e}} = \frac{s}{a} \quad (5.11)$$

where T_{1e} and T_{2e} are the effective stator and rotor turns per phase, and the turns ratio, a , between the stator-to-rotor effective turns per phase is given as

$$a = \frac{T_{1e}}{T_{2e}} = \frac{k_{w1}T_1}{k_{w2}T_2} = \text{turns ratio} \quad (5.12)$$

The rotor current I_{rr} , then, is

$$I_{rr} = \frac{E_2}{R_{rr} + j\omega_s L_{lrr}} = \frac{E_2}{R_{rr} + j\omega_s L_{lrr}} \quad (5.13)$$

Substituting for E_2 from equation (5.11) into equation (5.13), the rotor current is

$$I_{rr} = \frac{E_1}{\frac{aR_{rr}}{s} + j\omega_s(aL_{lrr})} = \frac{E_1/a}{\frac{R_{rr}}{s} + j\omega_s L_{lrr}} \quad (5.14)$$

Equation (5.14) is incorporated into the equivalent circuit as shown in Figure 5.2. Note that both the rotor and stator uniformly have the same frequency, which is that of the stator in equation (5.14). The rotor current reflected into the stator is denoted as I_r in Figure 5.2 and is given in terms of the rotor current I_{rr} as

$$I_r = \frac{I_{rr}}{a} \quad (5.15)$$

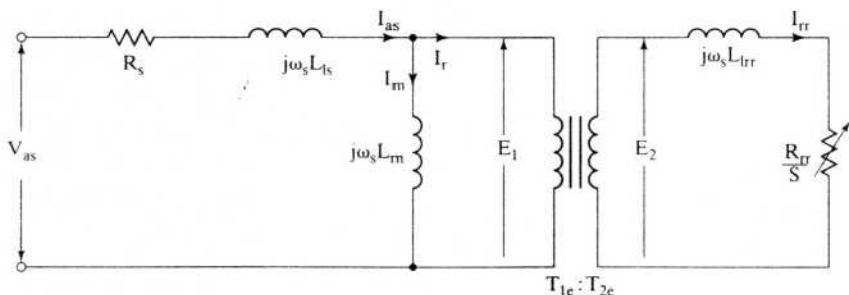


Figure 5.2 Equivalent circuit of the induction motor

CHAPTER 6

Phase-Controlled Induction-Motor Drives

6.1 INTRODUCTION

The speed of the induction motor in terms of slip, frequency, and pole numbers is written as

$$\omega_m = \frac{\omega_r}{P/2} = \frac{2}{P} \cdot \omega_s (1 - s) = \frac{4\pi}{P} \cdot f_s (1 - s) \quad (6.1)$$

The rotor speed of the induction motor can be varied by changing the number of poles, the slip, or the supply frequency. Pole-amplitude-modulated motors operate by changing the pole numbers, which in turn, is achieved with a change of winding connections. They require relays and circuit breakers to change winding connections. They provide a very limited but stepped (2 or 3) form of speed control, because the supply frequency is constant. They are not prevalent in practice nowadays.

The *slip* speed control is effected through the variation of applied voltage or insertion of external resistors in rotor or stator. They take different forms in implementation. As the maximum efficiency is $(1 - s)$, efficiency is poor at low speeds with slip-controlled drive systems. In this chapter, slip control with stator voltage control and slip energy-recovery control are considered for study. These drives are studied with special emphasis on the converters used, control strategy, steady-state performance analysis, harmonics and their impact on the performance and rating of the motors and converters for various applications, starting provisions, feedback control, and modern developments. Speed control with frequency variation is dealt with in Chapters 7 and 8.

6.2 STATOR VOLTAGE CONTROL

6.2.1 Power Circuit and Gating

By having series-connected power switches in the induction motor, the instant of voltage application can be delayed. This is accomplished by controlling the gating / base drive signals to the power switches. A power-circuit configuration of stator-phase control methods is shown in Figure 6.1. The power switches can be SCRs, triacs, power transistors, or GTOs. The power switches are numbered to reflect the sequence of their gating control. This is very similar to the control of a three-phase controlled-bridge converter discussed in Chapter 3.

The gating signals are synchronized to the phase voltages, and they can be delayed up to a maximum of 180 degrees. The angle of delay, α , is termed the triggering angle. The gating signals are spaced at 60 degrees interval from each other. For sustained conduction, the gating signals are more than 60 degrees wide so that two power devices in two different phases conduct at a given time for current flow. The load is inductive, and the gating signals need to be maintained for the entire conduction time to ensure continued conduction (or until the current is higher than the latching current of the SCR).

6.2.2 Reversible Controller

For the phase controller shown in Figure 6.1, the power can flow only *from* the 3-phase supply *to* the machine, and it can run in only one direction for only one possible phase sequence of the output voltage. Thus, it has only one-quadrant torque-speed

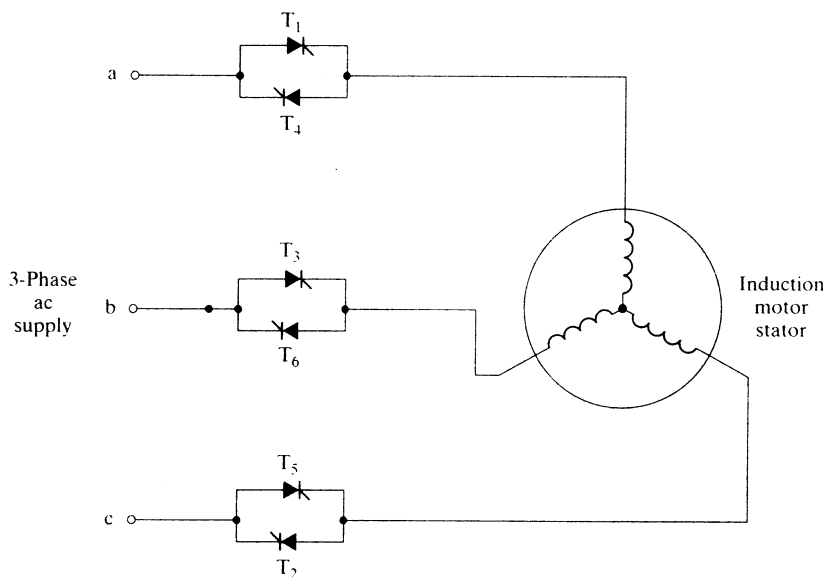


Figure 6.1 Phase-controlled induction-motor drive

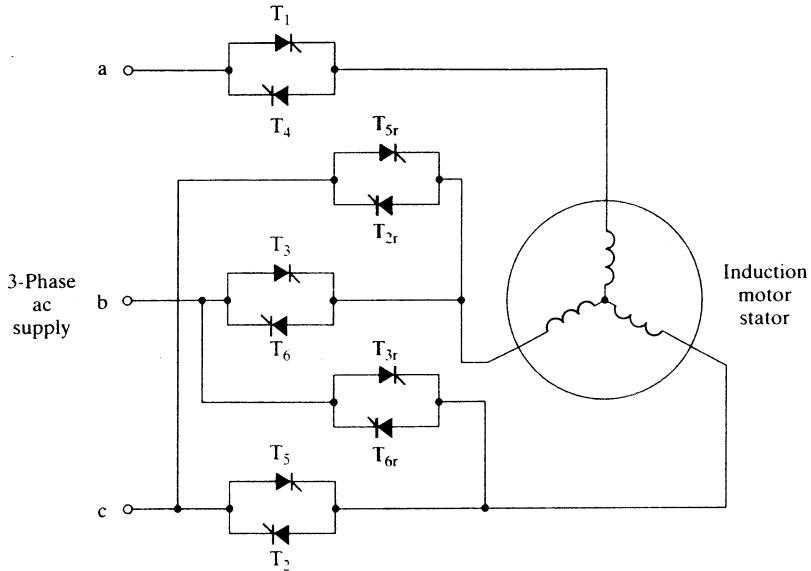


Figure 6.2 Reversible phase-controlled induction-motor drive

performance. To make this motor run in the other direction, the phase sequence of the input supply voltage is reversed. Two terminals of the stator windings need to be interchanged for phase-sequence reversal. It is accomplished with the addition of two pairs of antiparallel power switches, as shown in Figure 6.2. These power devices are denoted by an additional subscript, r . The sequence of gating for one direction of rotation (say, clockwise) is $T_1 T_2 T_3 T_4 T_5 T_6$; the sequence of gating for counterclockwise rotation is $T_1 T_{2r} T_{3r} T_4 T_{5r} T_{6r}$. During the counterclockwise rotation, note that the devices T_2 , T_3 , T_5 , and T_6 are not gated and hence remain open. By gating their counterparts, the phase sequence has been changed, to change the rotational direction of the motor.

The reversible phase-controlled induction-motor drive operates in the first and third quadrants of the torque–speed characteristics. It cannot operate in the II and IV quadrants and avoids regeneration. In avoiding regeneration, this motor drive becomes slow in going from one speed in one direction to another speed in the opposite direction. This is explained by using Figure 6.3.

Considering the present operating point, $P_1 (\omega_{m1}, T_{e1})$, and the required point to be reached, $P_2 (-\omega_{m2}, -T_{e2})$, the motor has to be slowed down to zero speed if stator currents are not allowed to exceed the rated values. The triggering angle is delayed so as to produce zero torque, and then the load slows down the rotor. This is shown in the diagram from P_1 to A and then to 0, respectively. At zero speed, the phase sequence is changed and the triggering angle is retarded until it can produce currents to generate the required torque trajectory, $-T_{e2}$, shown from 0 to B. Maintaining the torque at $-T_{e2}$ accelerates the rotor and load to P_2 , shown as a line from B to P_2 . It is assumed that T_{e1} and $-T_{e2}$ are the maximum allowable torques in their respective directions of rotation.

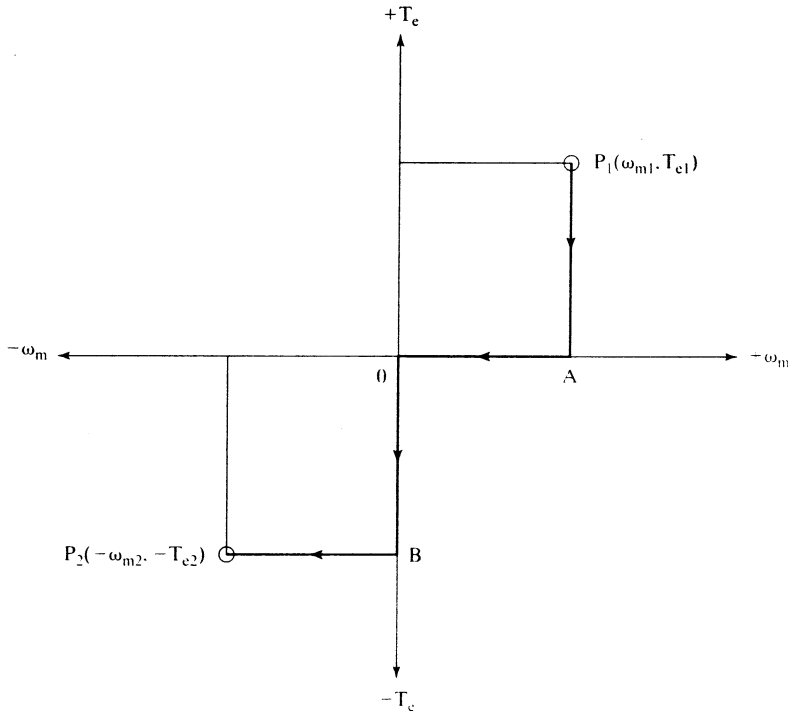


Figure 6.3 Quadrant change in the reversible phase-controlled induction-motor drive

If faster speed reversal of the motor is required, then the phase sequence of the supply is changed at P_1 instead of at 0. Then the induction motor is running opposite to the rotating magnetic field. This results in a slip greater than one, keeping the motor operation in the braking region. This would slow down the motor much faster than the load alone, but it is necessary to maintain the stator currents within safe levels, which is made possible by advancing the triggering angle. Even then, there will be excessive currents at the beginning of the phase sequence change, and they will last for a very short time. The current loop will force the triggering angle to advance to an appropriate value to keep the current below the safe level.

6.2.3 Steady-State Analysis

The torque–speed characteristics of the phase-controlled induction motor are required to evaluate its suitability for a load. The only control input is the triggering angle, α . Consider simplified but conceptual voltage and current waveforms for a phase shown in Figure 6.4. When there is no current in the stator, for instance between $(\alpha + \beta)$ and $(\pi + \alpha)$, the induced emf in the winding is reflected onto the phase terminal voltage. This induced emf is due to the air gap flux linkage, which is the resultant of stator and

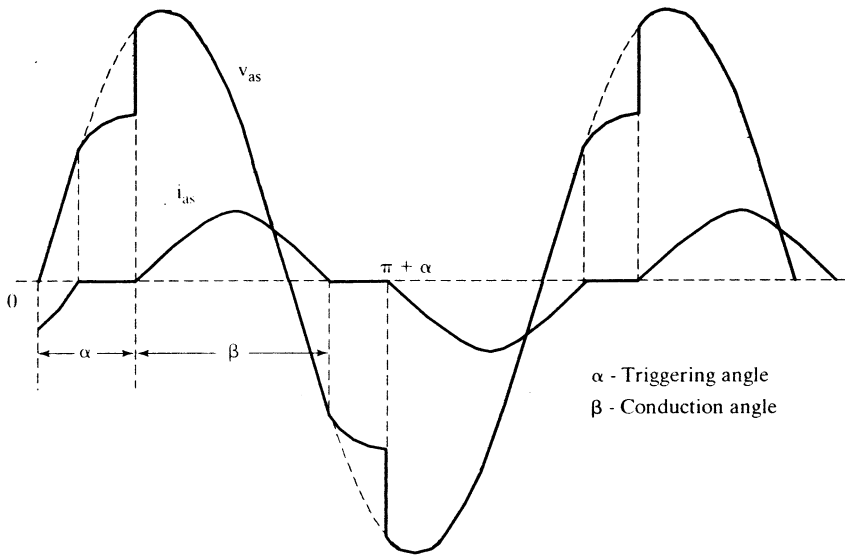


Figure 6.4 Phase voltage and current waveforms

rotor flux linkages. Due to the inductance in the rotor winding, even when stator current goes to zero, the rotor current continues to flow but decays exponentially. That rotor current contributes to the rotor and hence air gap flux linkages, resulting in the induced emf. The induced emf due to the air gap field will be sinusoidal, because the stator winding is sinusoidally distributed. Note that this reflected induced emf will be lower than the applied voltage. For a specified triggering angle, the determination of the conduction angle β is cumbersome, as it is an implicit function of the above parameters. A number of methods have been developed to compute β (and hence the steady-state currents and torque) by using state-variable techniques. They require the simulation of the motor in d and q frames and use of a computer, thus excluding a simple solution [1, 2, 3, 4]. Some simple analytical methods have come to the fore to give approximate results [2, 5, 6]. For most of the design and application studies, the accuracy of these methods is adequate. The analytical methods use the single-phase steady-state equivalent circuit of the induction motor. The performance of the induction motor with phase control is considered by one of the following methods:

- (i) The applied voltage is resolved into Fourier components and their individual current responses are added to compute the resultant current [5].
- (ii) The conduction angle is calculated by using the machine-equivalent circuit, and then the current is calculated [2].
- (iii) A closed-form solution approach [5] is used.

Method (ii) is considered here because of its simplicity; even though method (iii) is elegant, it requires significant computational effort. Method (i) requires consideration of a large number of harmonic equivalent circuits, with higher computational burden.

6.2.4 Approximate Analysis

6.2.4.1 Motor model and conduction angle. The induction motor is modeled as an equivalent resistance, R_{im} , in series with an equivalent reactance, X_{im} , by using the equivalent circuit. These equivalent parameters are written in terms of the motor parameters and slip as

$$R_{im} = R_s + \frac{X_m^2}{\left(\frac{R_r}{s}\right)^2 + (X_m + X_{lr})^2} \cdot \frac{R_r}{s} \quad (6.2)$$

$$X_{im} = X_{ls} + \frac{\left(\frac{R_r}{s}\right)^2 + X_{lr}(X_m + X_{lr})}{\left(\frac{R_r}{s}\right)^2 + (X_m + X_{lr})^2} \cdot X_m \quad (6.3)$$

The voltage applied to this equivalent circuit consisting of R_{im} and X_{im} is

$$v(t) = V_m \sin(\omega_s t + \alpha) \quad (6.4)$$

as shown in Figure 6.4, where α is the triggering delay angle.

The stator current is then

$$i_{as}(t) = \frac{V_m}{Z_{im}} [\sin(\omega_s t + \alpha - \phi) - \sin(\alpha - \phi)e^{-\frac{\omega_s t}{\tan \phi}}], \text{ for } 0 \leq \omega_s t \leq \beta \quad (6.5)$$

where the induction-machine equivalent impedance and power-factor angle are

$$Z_{im} = \sqrt{R_{im}^2 + X_{im}^2} \quad (6.6)$$

$$\phi = \tan^{-1} \left(\frac{X_{im}}{R_{im}} \right) \quad (6.7)$$

The conduction angle, β , is obtained from equation (6.5) when the current becomes zero. Such a condition is denoted by

$$i_{as} \left(\frac{\beta}{\omega_s} \right) = 0 \quad (6.8)$$

and hence,

$$\sin(\beta + \alpha - \phi) - \sin(\alpha - \phi)e^{-\left(\frac{\beta}{\tan \phi}\right)} = 0 \quad (6.9)$$

The flowchart is given in Figure 6.5 to solve the equation (6.9), using the Newton–Raphson technique. When the triggering angle is less than the power-factor angle, the current will conduct for the positive half-cycle, say from α to $(\pi + \phi)$. When the negative half-cycle is applied at $(\pi + \alpha)$, and as $\alpha < \phi$, still the positive current is flowing. Note that the voltage applied to the machine is negative. As the current goes to zero, the negative half-cycle has already been activated by triggering at $(\pi + \alpha)$. Therefore, the machine receives full source voltage, and hence the current-

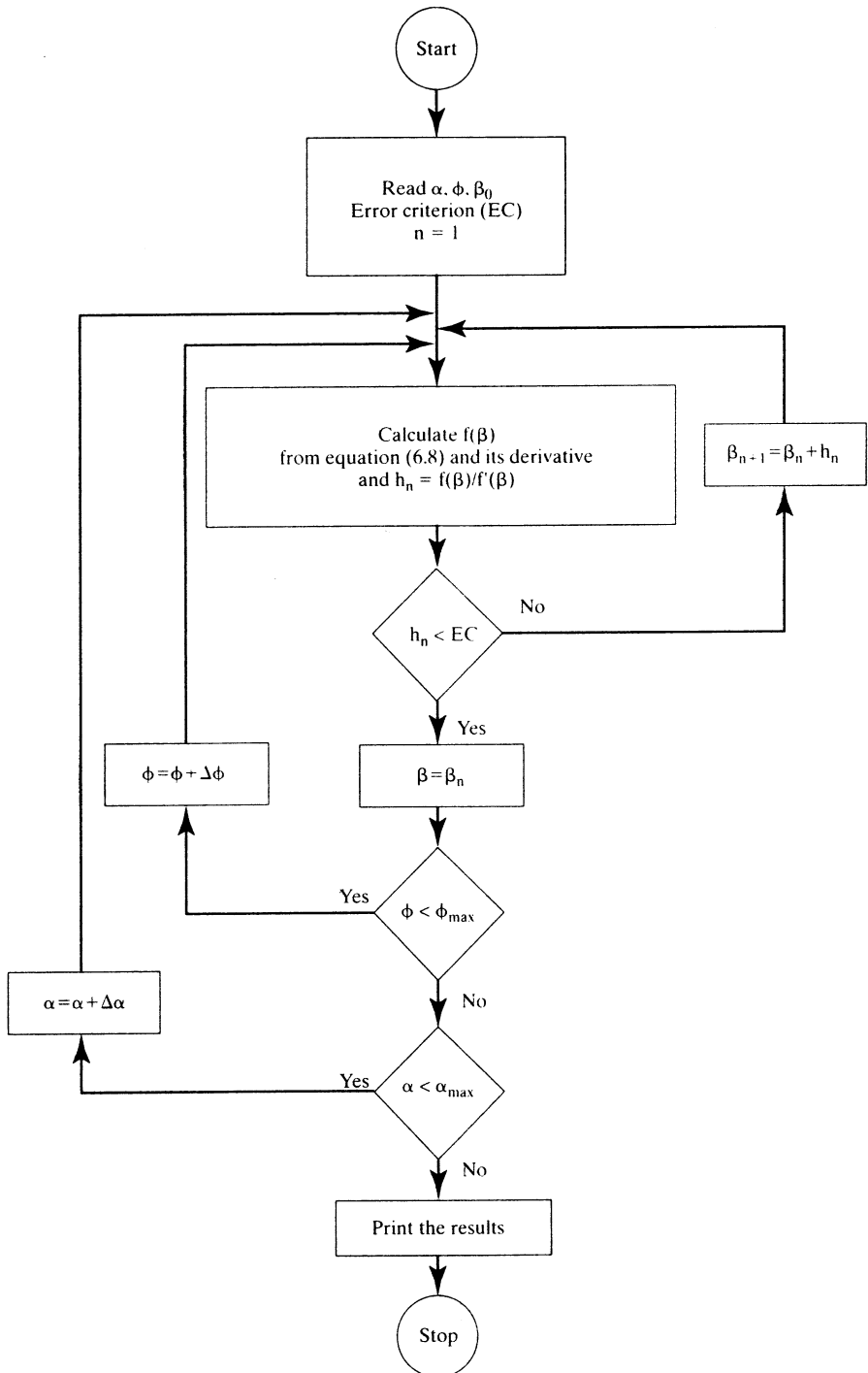


Figure 6.5 Flowchart for the computation of triggering angle vs. conduction angle as a function of power-

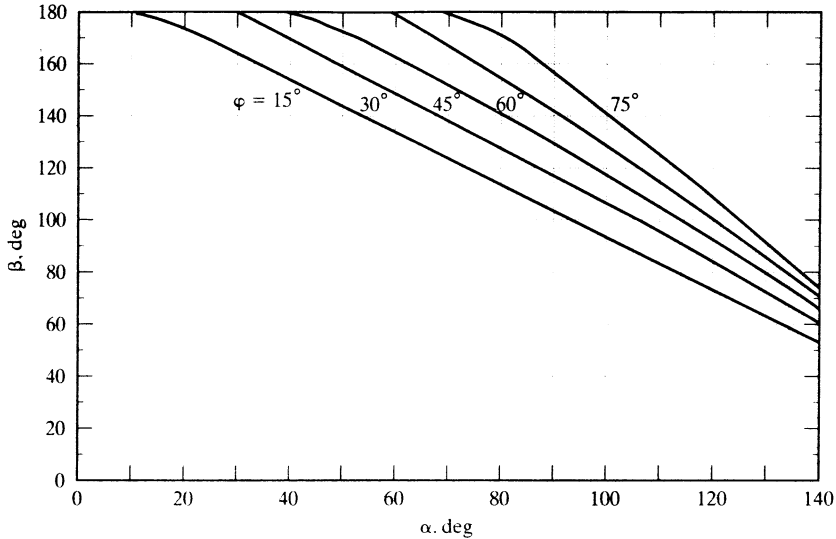


Figure 6.6 Conduction angle vs. triggering angle for various power-factor angles

conduction angle is 180 degrees. For the case when $\alpha < \phi$, the condition that $\beta = 180$ degrees has to be included in the solution of equation (6.9).

The solution of the transcendental equation (6.9) gives the value of β , and the instantaneous current $i_{as}(t)$ is evaluated from the equation (6.5). The relationship between the conduction angle and triggering angle for power-factor angles of 15° , 30° , 45° , 60° , and 75° is shown in Figure 6.6. Table 6.1 contains the relationship between ϕ and β for various α , for easy reference.

The stator voltage has to be resolved into its Fourier components, from which the fundamental current and torque developed in the motor are computed. The resolution of the stator current is derived in normalized form so that the results can be generalized.

Let the normalizing or base variables be

$$V_b = \frac{V_m}{\sqrt{2}} = \text{Rated rms phase voltage of the motor} \quad (6.10)$$

where V_m is the peak-phase voltage

$$I_b = \text{Rated rms current of the machine} \quad (6.11)$$

$$Z_b = \frac{V_b}{I_b} \quad (6.12)$$

6.2.4.2 Fourier resolution of voltage. The fundamental of the input-phase voltage is derived to be

$$v_{a1} = \sqrt{(a_1^2 + b_1^2)} \sin(\omega_s t + \theta_1) \quad (6.13)$$

TABLE 6.1 ϕ vs. β for various triggering angles

α , deg	10	20	30	40	50	60	70	80	90	100	110	120	130	140
ϕ , deg	β , deg													
9	179	169.00	159.00	149.00	139.00	129.00	119.00	109.00	99.00	89.00	78.99	68.97	58.93	48.80
12	180	172.00	162.00	152.00	142.00	132.00	122.00	111.99	101.99	91.97	81.93	71.85	61.68	51.33
15	180	175.00	165.00	155.00	145.00	134.99	124.99	114.97	104.94	94.88	84.77	74.57	64.21	53.57
18	180	178.00	168.00	158.00	147.99	137.98	127.95	117.91	107.83	97.70	87.48	77.11	66.51	55.54
21	180	180.00	171.00	160.99	150.97	140.94	130.89	120.80	110.65	100.41	90.05	79.47	68.61	57.29
24	180	180.00	173.99	163.97	153.94	143.88	133.78	123.63	113.39	103.02	92.48	81.68	70.53	58.87
27	180	180.00	176.99	166.96	156.90	146.80	136.64	126.40	116.04	105.52	94.79	83.75	72.30	60.30
30	180	180.00	180.00	169.94	159.84	149.69	139.46	129.11	118.63	107.94	96.99	85.70	73.96	61.61
33	180	180.00	180.00	172.93	162.79	152.57	142.25	131.79	121.15	110.28	99.11	87.55	75.50	62.82
36	180	180.00	180.00	175.94	165.74	155.44	145.02	134.42	123.62	112.55	101.14	89.32	76.97	63.95
39	180	180.00	180.00	178.98	168.71	158.32	147.78	137.04	126.06	114.77	103.12	91.02	78.36	65.01
42	180	180.00	180.00	180.00	171.71	161.22	150.55	139.64	128.47	116.96	105.05	92.66	79.69	66.01
45	180	180.00	180.00	180.00	174.76	164.16	153.33	142.25	130.87	119.11	106.94	94.26	80.97	66.97
48	180	180.00	180.00	180.00	177.88	167.14	156.15	144.88	133.27	121.27	108.81	95.82	82.21	67.89
51	180	180.00	180.00	180.00	180.00	170.19	159.03	147.55	135.69	123.42	110.67	97.36	83.43	68.77
54	180	180.00	180.00	180.00	180.00	173.34	161.97	150.26	138.15	125.59	112.52	98.89	84.62	69.64
57	180	180.00	180.00	180.00	180.00	176.59	165.01	153.05	140.65	127.78	114.38	100.41	85.80	70.48
60	180	180.00	180.00	180.00	180.00	180.00	168.17	155.92	143.22	130.01	116.27	101.94	86.97	71.31
63	180	180.00	180.00	180.00	180.00	180.00	171.48	158.92	145.87	132.31	118.19	103.48	88.14	72.14
66	180	180.00	180.00	180.00	180.00	180.00	174.97	162.06	148.64	134.68	120.15	105.04	89.32	72.95
69	180	180.00	180.00	180.00	180.00	180.00	178.70	165.39	151.54	137.14	122.18	106.64	90.52	73.78
72	180	180.00	180.00	180.00	180.00	180.00	180.00	168.94	154.61	139.73	124.29	108.29	91.73	74.60
75	180	180.00	180.00	180.00	180.00	180.00	180.00	172.77	157.90	142.46	126.49	109.99	92.97	75.44
78	180	180.00	180.00	180.00	180.00	180.00	180.00	176.96	161.44	145.38	128.82	111.77	94.26	76.30
81	180	180.00	180.00	180.00	180.00	180.00	180.00	180.00	165.32	148.53	131.29	113.63	95.59	77.17
84	180	180.00	180.00	180.00	180.00	180.00	180.00	180.00	169.61	151.96	133.95	115.61	96.98	78.08
87	180	180.00	180.00	180.00	180.00	180.00	180.00	180.00	174.44	155.75	136.83	117.72	98.44	79.02
90	180	180.00	180.00	180.00	180.00	180.00	180.00	180.00	180.00	160.00	140.00	120.00	100.00	90.00

where

$$a_1 = [\cos 2\alpha - \cos(2\alpha + 2\beta)] \frac{V_m}{2\pi} \quad (6.14)$$

$$b_1 = [2\beta + \sin 2\alpha - \sin(2\alpha + 2\beta)] \frac{V_m}{2\pi} \quad (6.15)$$

$$\theta_1 = \tan^{-1} \left(\frac{a_1}{b_1} \right) \quad (6.16)$$

In terms of normalized variables, the rms phase voltage is given by

$$V_{an} = \frac{V_{a1}}{V_b} = \frac{1}{\pi \sqrt{2}} [1 + 2\beta^2 + 2\beta \{\sin 2\alpha - \sin(2\alpha + 2\beta)\} - \cos 2\beta]^{1/2} \text{ p.u.} \quad (6.17)$$

A set of normalized curves between V_{an} and β for various triggering angles is shown in Figure 6.7.

6.2.4.3 Normalized currents. The normalized fundamental of the phase current is obtained from

$$I_{an} = \frac{V_{an}}{Z_{an}} = \frac{V_{an}}{(Z_{im}/Z_b)} \cdot \text{p.u.} \quad (6.18)$$

The normalized rotor current is derived as

$$I_m = \frac{X_m}{\sqrt{\left(\frac{R_r}{s}\right)^2 + (X_m + X_{lr})^2}} \cdot I_{an} \text{ p.u.} \quad (6.19)$$

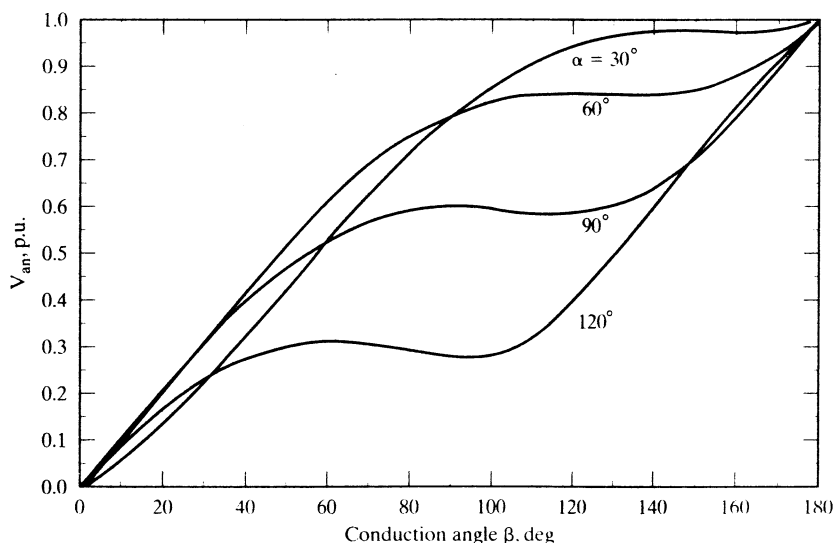


Figure 6.7 Normalized fundamental output voltage vs. conduction angle for various triggering-delay angles

6.2.4.4 Steady-state performance computation. The steady-state performance in terms of torque–speed characteristics is evaluated as in the following:

- Step 1:** Assume a slip and, using the equivalent circuit parameters, compute Z_{an} and ϕ .
- Step 2:** Given α and, from the computed power-factor angle, ϕ , read the conduction angle from the normalized graph shown in Figure 6.6 or Table 6.1.
- Step 3:** From β , the fundamental applied voltage is evaluated by using equation (6.17) or from Figure 6.7.
- Step 4:** The stator current is calculated from V_{an} and Z_{an} , and then the rotor current can be calculated from stator current and hence also the normalized torque.
- Step 5:** For various values of slip and α , I_{an} and T_{en} are evaluated and torque–speed characteristics are drawn.

Step 4 needs the expression to calculate the normalized electromagnetic torque, which is derived as

$$\begin{aligned} T_e &= 3 \frac{P}{2} \cdot \frac{I_r^2 R_r}{s \omega_s} = 3 \frac{P}{2} \left(\frac{I_r}{I_b} \right)^2 \cdot \frac{I_b R_r}{V_b s \omega_s} \cdot I_b \cdot V_b = \left(3 \frac{P}{2 \omega_s} \cdot V_b I_b \right) I_m^2 \cdot \frac{R_m}{s} \quad (6.20) \\ &= \frac{3 V_b I_b}{\left\{ \frac{\omega_s}{(P/2)} \right\}} \cdot I_m^2 \cdot \frac{R_m}{s} \end{aligned}$$

Noting that

$$\text{Base torque} = T_b = \frac{3 V_b I_b}{\left\{ \frac{\omega_s}{(P/2)} \right\}} \quad (6.21)$$

$$\text{Base power} = P_b = 3 V_b I_b \quad (6.22)$$

and dividing equation (6.20) by (6.21) yields the normalized torque as

$$\frac{T_e}{T_b} = T_{en} = I_m^2 \cdot \frac{R_m}{s}, \text{ p.u.} \quad (6.23)$$

where the normalized rotor resistance is

$$R_{rn} = \frac{I_b R_r}{V_b} \quad (6.24)$$

I_{rn} and R_{rn} are the normalized rotor current and resistance, respectively. The mechanical power output is

$$P_{on} = I_{rn}^2 \frac{R_{rn}}{s} (1 - s) \quad (6.25)$$